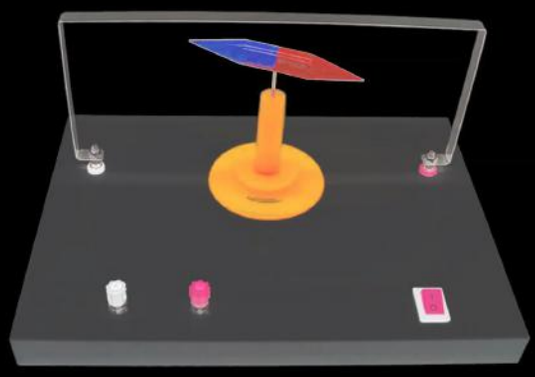
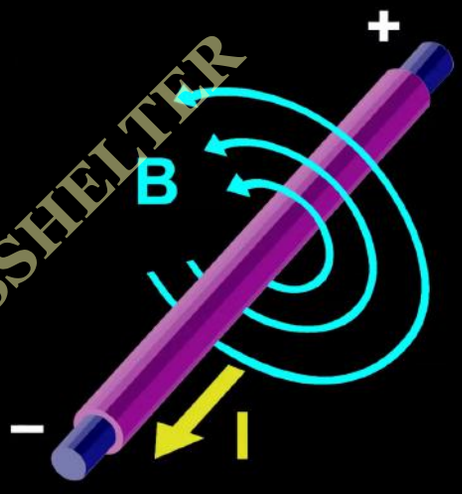
MAGNETIC FIELD: Just as stationary charges produce an electrostatic field which is responsible for electrostatic interaction forces, moving charges produce a magnetic field which is responsible for magnetic interaction force with other charged particles. Magnetic materials such as lodestone have been known to man since very ancient times. At one point of time, electricity and magnetism were thought to be independent phenomena, however Oersted observed that a current carrying wire could deflect a magnetic compass needle and that initiated a quest for finding the link between electricity and magnetism. Now it is well understood that charges in motion give rise to magnetic fields. During our study of magnetism, we will be first focusing on effects of magnetic field (i.e forces and torques due to magnetic fields) and later on we will study about generation of magnetic fields. Unit of magnetic field is Tesla (T) or Weber/m^2 (Wb/m^2).



OERSTED'S EXPERIMENT



MAGNETIC FIELD DUE TO CURRENT CARRYING WIRE

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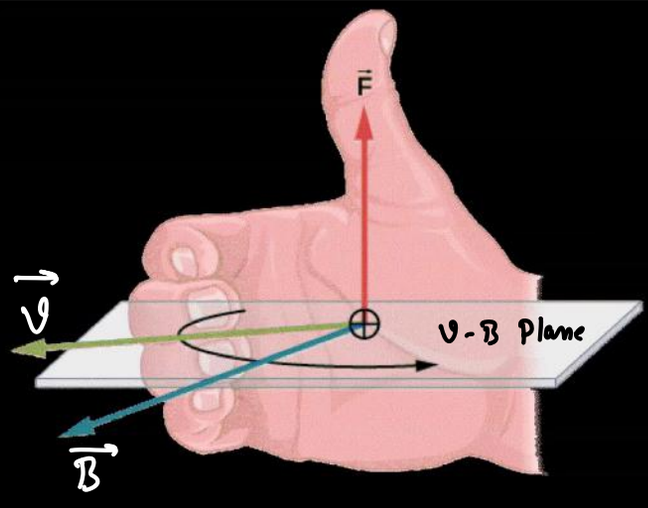
MAGNETIC FORCE ON A MOVING CHARGE:

Let q be the charge of a particle moving with a velocity $\vec{v}$ in a region where magnetic field is $\vec{B}$ then the magnetic force on the particle is given as:

$$\boxed{\vec{F} = q\,\vec{v} \times \vec{B}}$$



→ A particle at rest experiences no magnetic force

→ A particle moving || to magnetic field experiences no magnetic force.

→ $\vec{F} \perp \vec{v}$ thus $\vec{F} \cdot \vec{v} = 0$ thus magnetic force does no work on a charged particle and hence cannot change its kinetic energy.

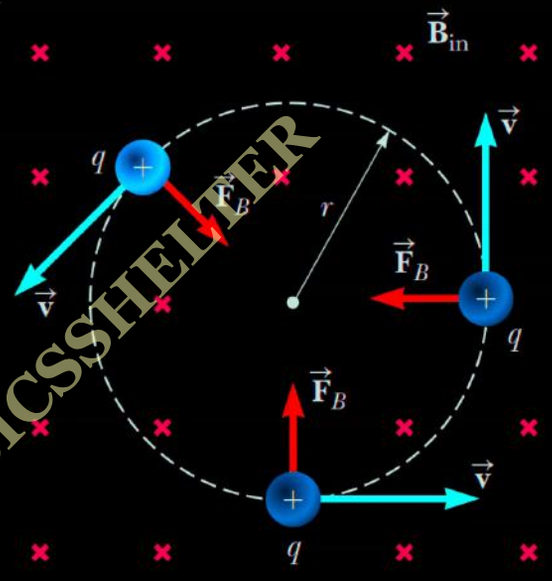
→ This means that a particle experiencing only magnetic force has zero tangential acceleration, it has only centripetal acceleration such that $m(\vec{\omega} \times \vec{v}) = q\vec{v} \times \vec{B}$

Motion Of Charged Particle In Magnetic Field ($\vec{v} \perp \vec{B}$):

A particle launched in magnetic field with $\vec{v} \perp \vec{B}$ undergoes a circular motion. This can be demonstrated as follows:

$$F = qvB = mv^2/R$$

$$\Rightarrow \boxed{R = \frac{mv}{qB}}$$ now ∵ v is constant,



thus R must also be a constant, thus the motion of particle is circular.

→ $\omega = \frac{v}{R} \Rightarrow \boxed{\omega = \frac{qB}{m}}$

→ $T = 2\pi/\omega \Rightarrow \boxed{T = \frac{2\pi m}{qB}}$

→ $\nu = 1/T \Rightarrow \boxed{\nu = \frac{qB}{2\pi m}}$

→ Sign of charge decides sense of rotation. Sense of rotation of the particle can be quickly found as follows: For an electron, point the thumb of right hand along the magnetic field, now the curl of fingers gives the sense of rotation of the electron (or negatively charged particles). For protons (or +Ve particles) the sense of rotation is just opposite of this.

Example: A charged particle is projected in a magnetic field $\vec{B} = (3\hat{i} + 4\hat{j}) \times 10^{-2}\,T$, the acceleration of particle is found to be $\vec{a} = (x\hat{i} + 2\hat{j})\,m/s^2$. Find the value of x.

Solution: ∵ $\vec{F} \perp \vec{B}$ thus $\vec{a} \perp \vec{B}$ thus $\vec{a} \cdot \vec{B} = 0 \Rightarrow 3x + 8 = 0$

$\Rightarrow x = -8/3$ Ans.

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Specific Charge (α)

The ratio of charge to mass of any particle is called the specific charge (α). Thus $\alpha = q/m$. The angular speed of any particle can be expressed as $\omega = qB/m = \alpha B$. Thus particles having same specific charge will have same angular frequency (and hence time period of revolution) e.g. Deutron (^2_1H) and an α particle (^4_2He) have same time periods of revolution.

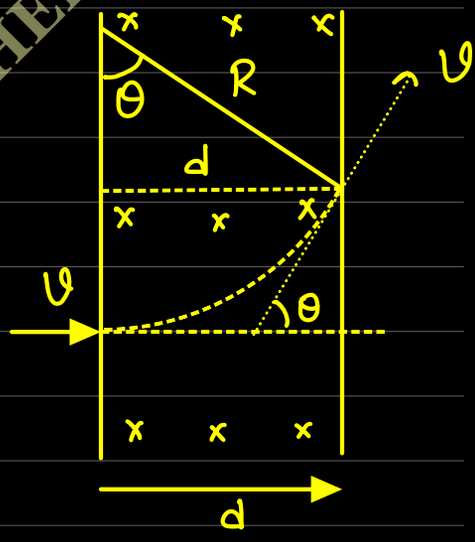
→ Period of revolution is independent of projection speed.

→ $R = mv/qB = v/\alpha B$, thus speed determines radius (and not time period).

Angle Of Deviation Of Charged Particle Through A Strip (v ⊥ Boundary)

Consider a charged particle entering a strip of uniform magnetic field as shown in adjacent figure, its angle of deviation is given as:

$$\sin\theta = d/R \Rightarrow \boxed{\theta = \sin^{-1}(d/R)}$$



→ If $d > R$, the particle makes a full 180° turn. (i.e a semicircular U turn).

→ If $d \lesssim R$, $\theta \approx 90°$.

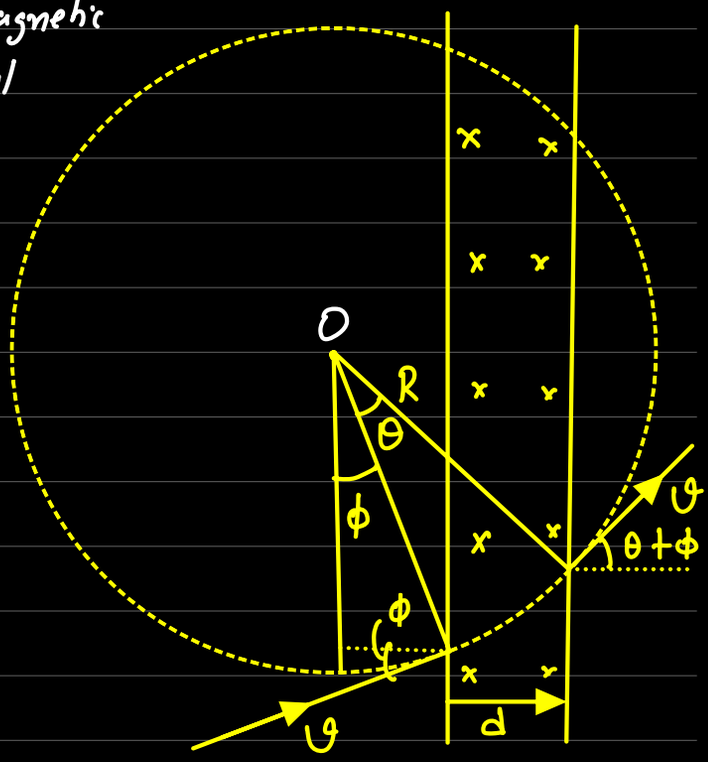
Angle Of Deviation Through A Magnetic Strip (v not ⊥ to Boundary):

Consider a particle entering a strip of magnetic field at an angle ϕ above horizontal as shown in the figure. For this, angle of deviation θ can be worked out using a little geometry or trigonometry. It can be readily seen that:

$$R\sin(\theta+\phi) - R\sin\phi = d$$

Thus,

$$\theta = \sin^{-1}\left[\left(\frac{d}{R} + \sin\phi\right)\right] - \phi$$



→ The key idea here is to visualise the complete circle and then work out the

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trigonometric relation. (Always draw a figure before writing the equations).

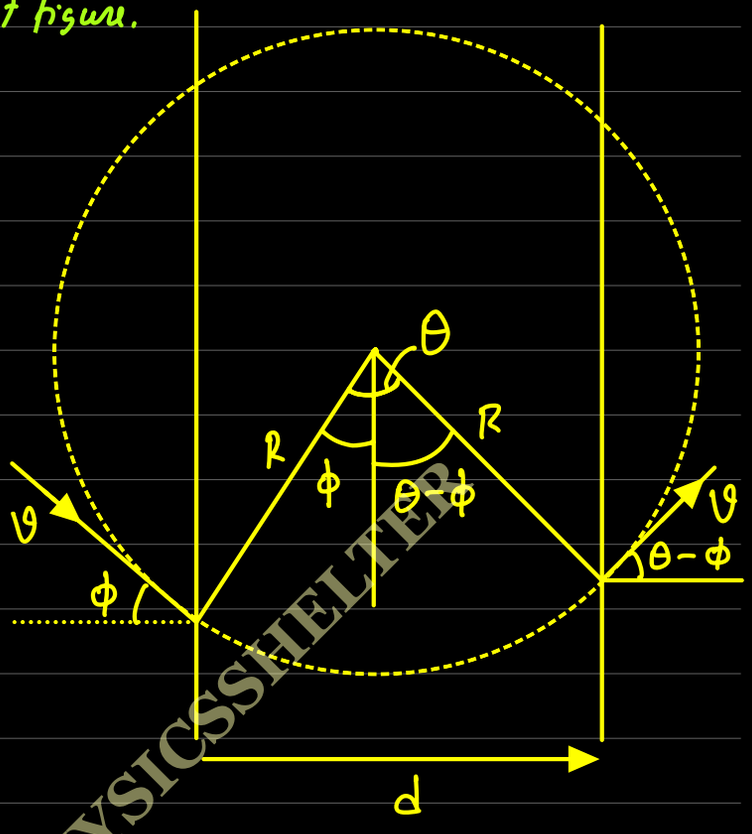
→ The centre of circle may lie inside or outside the region of magnetic field. See the possibility in adjacent figure.

Here:

$$R\sin\phi + R\sin(\theta-\phi) = d$$

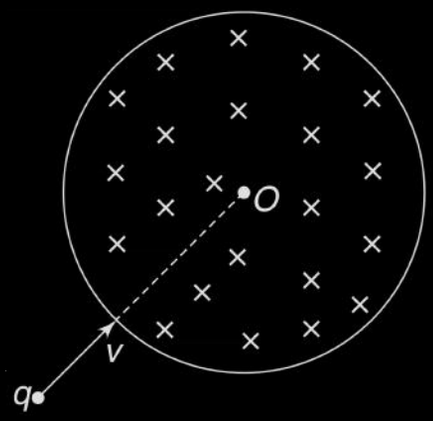
$$\Rightarrow \theta = \sin^{-1}\left(\frac{d}{R} - \sin\phi\right) + \phi$$

→ If the argument of $\sin^{-1}$ term is greater than 1 then that means particle will not be able to cross the magnetic strip and emerge from same side of the strip



EXAMPLE: SB TIWARI - 15:

Figure shows a circular region of radius $R = \sqrt{3}\,m$ which has a uniform magnetic field $B = 0.2\ T$ directed into the plane of the figure. A particle having mass $m = 2$ g, speed $v = 0.3$ m/s and charge $q = 1$ mC is projected along the radius of the circular region as shown in figure. Calculate the angular deviation produced in the path of the particle as it comes out of the magnetic field. Neglect any other force apart from the magnetic force.



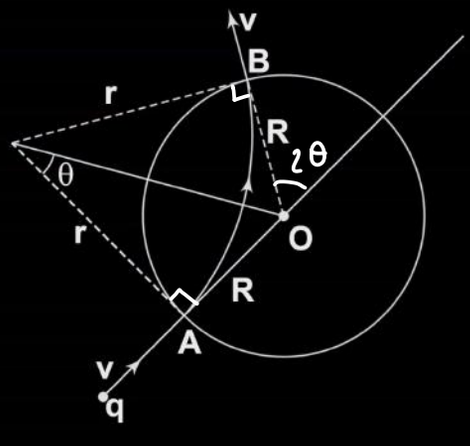
Solution: Consider the adjacent figure:

Angle of deviation = 2θ.

$$\tan\theta = R/s = R/(mv/qB) = qBR/mv$$

Thus $\theta = \tan^{-1}(qBR/mv)$ ⇒

Angle of deviation = $2\theta = 2\tan^{-1}(qBR/mv)$



$$\Rightarrow 2\theta = 2\tan^{-1}\frac{10^{-3}\times 0.2\times\sqrt{3}}{2\times10^{-3}\times 0.3} = 2\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

⇒ Deviation = $2\theta = 60°$ Ans.

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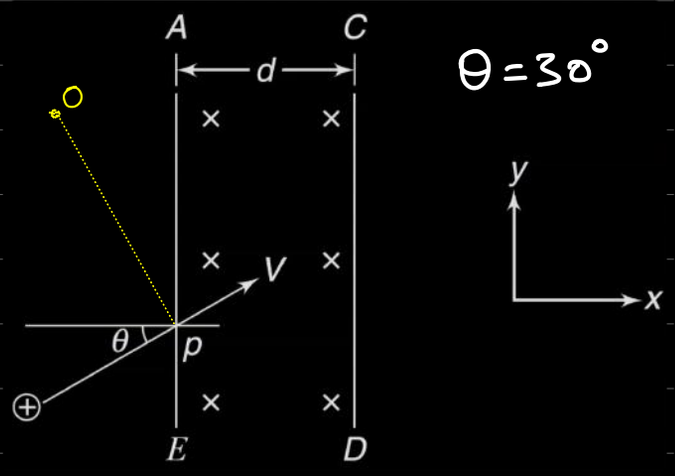
EXAMPLE: Find the deviation in path of the particle if $d = \frac{5mv}{4qB}(1-\sin\theta)$.



Solution: Assuming that the particle does emerge on other side, let the angle of deviation be ϕ. This is the case where centre of curvature lies outside the region. Thus:

$$R\sin(\theta+\phi) - R\sin\theta = d$$

$$\Rightarrow \phi = \sin^{-1}\left(\frac{d}{R} + \sin\theta\right) - \theta \quad , \text{ Now } R = \frac{mv}{qB} \Rightarrow \frac{d}{R} = \frac{5}{4}(1-\sin\theta)$$

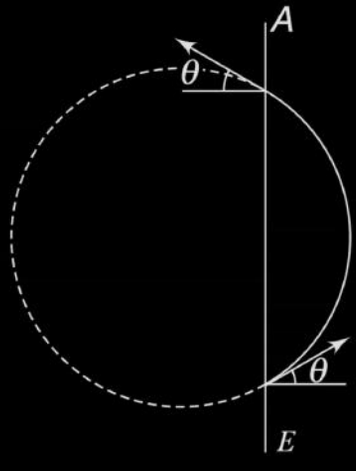
$$\Rightarrow \phi = \sin^{-1}\left(\frac{5}{4}(1-\sin\theta) + \sin\theta\right) - \theta$$

$$\Rightarrow \phi = \sin^{-1}\left[\frac{5}{4} - \frac{1}{4}\sin\theta\right] - \theta$$

Now if $\theta = 30^\circ$, $\Rightarrow \phi = \sin^{-1}\left[\frac{5}{4} - \frac{1}{8}\right] - 30^\circ$, Here the expression in parenthesis is not defined thus our assumption is not correct. Therefore the particle must emerge on the same side.

Now we can easily find the angle of deviation. Consider the adjacent figure, depicting path of the particle. From the figure it can be seen that initial velocity vector makes θ with $+x$ axis and final velocity vector makes an angle $180^\circ-\theta$ with $+x$ direction. Thus angle of deviation : $\delta = (180^\circ-\theta) - \theta = 180^\circ - 2\theta$

i.e. 120° Ans.



Think: What are the permissible values of θ for emergence from other side?

Solution: For this argument of $\sin^{-1}$ should have a magnitude ≤ 1

$\Rightarrow \frac{5}{4} - \frac{1}{4}\sin\theta < 1 \Rightarrow \frac{1}{4}\sin\theta > \frac{1}{4} \Rightarrow \sin\theta > 1$, thus with given data it is impossible to emerge from other side. AMBARISH SIR

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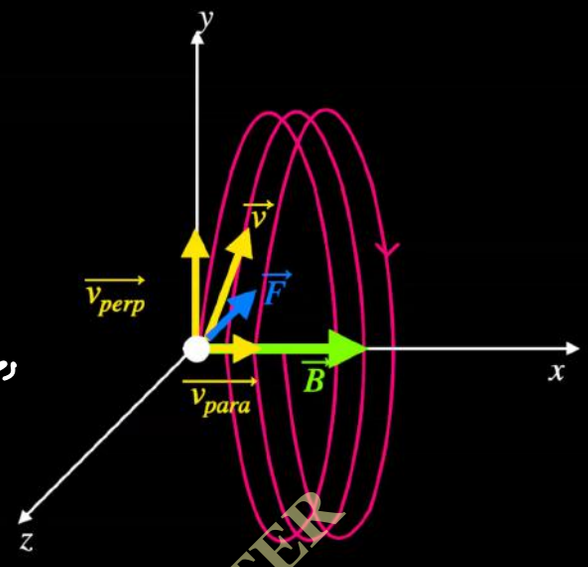
Projection In Uniform Magnetic Field $\vec{v}$ Not $\perp$ To $\vec{B}$:

If $\vec{v}$ is at some angle θ to $\vec{B}$ then we can resolve $\vec{v}$ in to components parallel and perpendicular to $\vec{B}$, i.e:

$$v_{\parallel} = v\cos\theta \quad \text{and}$$

$$v_{\perp} = v\sin\theta$$



Now because of $v_{\parallel}$ the particle experiences no force since $q\,\vec{v}_{\parallel} \times \vec{B} = 0$

Thus $\vec{v}_{\parallel}$ remains unchanged because the force is $\perp$ to $v_{\parallel}$ ($\because$ Force is $\perp$ to $\vec{B}$).

Now because of $v_{\perp}$, the particle has a tendency to execute circular motion. Thus overall effect is the execution of a helical motion. (In plane $\perp$ to B the projection is circle, but because of $v_{\parallel}$ the particle also keeps advancing along $\vec{B}$).

Larmor Radius (Gyro-Radius / Cyclotron Radius): As discussed in previous section, the charged particle executes a helical motion. The radius of enveloping cylinder of such a helix is called Larmor Radius. (This is also the radius of the projection circle in plane $\perp$ to B). This can be readily calculated as:

$$q\,v_{\perp}\,B = m\,v_{\perp}^2/R$$

$$\Rightarrow \boxed{R = \frac{m v_{\perp}}{qB}}$$

also $\boxed{R = \frac{mv\sin\theta}{qB}}$

Gyrofrequency: The angular frequency of the particle's projection $\perp$ to B is known as Gyrofrequency (in rad s^{-1}). This can be easily calculated as:

$$m\omega^2 R = q\,v_{\perp}\,B = m\omega v_{\perp} \Rightarrow$$

$$\boxed{\omega = \frac{qB}{m}} \quad \text{Thus} \quad \boxed{T = \frac{2\pi m}{qB}}$$

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AMBARISH SIR

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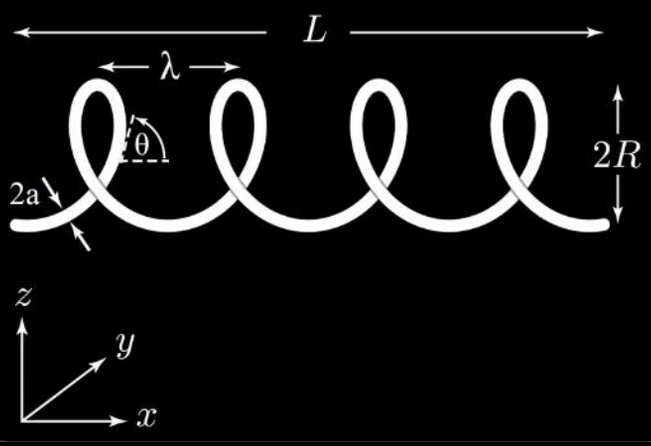
Pitch Of The Helix:

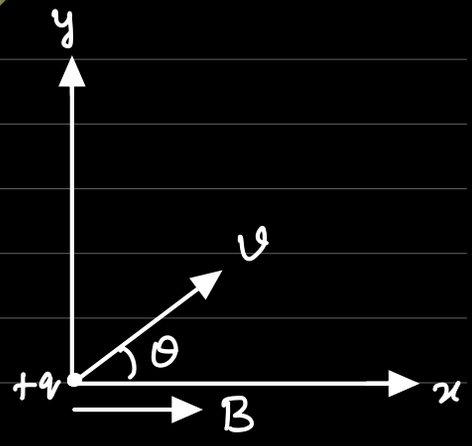
The distance through which the charged particle advances during one complete time period of revolution about axis of the helix is called the pitch (λ). [See adjacent figure]. Thus using definition, $\lambda = v_{\parallel} \times T$

$$\Rightarrow \boxed{\lambda = \frac{2\pi m v \cos\theta}{qB}}$$



Pitch Angle: The angle which the velocity vector makes with the magnetic field is called the pitch angle (θ). [See figure above]. $\tan\theta = v_{\perp} / v_{\parallel}$.

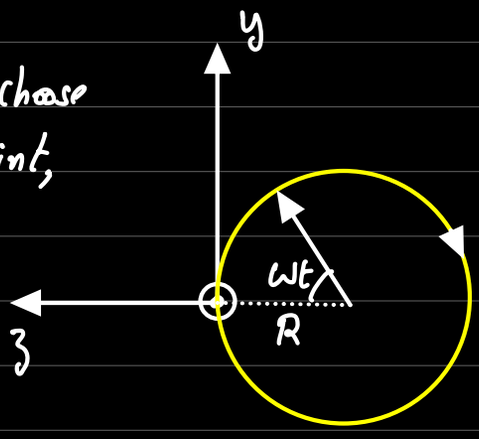
Example: A particle having charge q and mass m is launched from the origin with initial speed v in the xy plane in a region of uniform magnetic field $B\hat{i}$ (see figure). Find the position of the particle as a function of time.



Solution: $v_{\parallel} = v\cos\theta \Rightarrow x = v\cos\theta\, t$ ①

Now to visualise motion in the y-z plane we choose a view in which the x axis appears as a point, with x axis coming out of the page.

Now the sense of rotation for a positive charge is opposite to that for a negative charge (which is calculated using right hand thumb rule).



Now $R = mv/qB$, $\omega = qB/m$ thus:

$y = R\sin\omega t = \left(\frac{mv\sin\theta}{qB}\right)\sin(qBt/m)$ ②, $z = -R(1-\cos\omega t)$, thus,

$z = -\frac{mv\sin\theta}{qB}\left(1-\cos\frac{qBt}{m}\right)$ ③, Thus the coordinates are given by x, y and z respectively.

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Example: Irodov 3.383

A slightly divergent beam of non-relativistic charged particles accelerated by a potential difference V propagates from a point A along the axis of a straight solenoid. The beam is brought into focus at a distance l from the point A at two successive values of magnetic induction B_1 and B_2. Find the specific charge q/m of the particles.

Solution: For particles to reconverge at a distance l, the distance l must be an integer multiple of the pitch. Now,

$\lambda = \frac{2\pi m v \cos\theta}{qB}$, thus as B increases, λ decreases. Thus if at B_1, l is n^{th} multiple then at B_2 it should be $(n+1)$ multiple (assuming $B_2 > B_1$). Also $\cos\theta \approx 1$ because it is given that propagation is along axis of solenoid, with slight divergence. Thus:

$$l = \frac{2\pi m v}{qB_1} \times n = \frac{2\pi m v}{qB_2}(n+1)$$

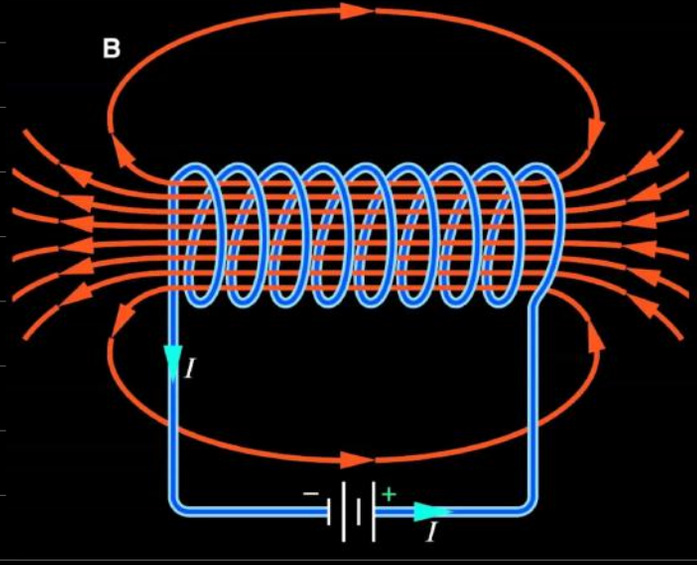
$$\Rightarrow n = \frac{q}{m}\frac{lB_1}{2\pi v} \quad \text{and} \quad n+1 = \frac{q}{m}\left(\frac{lB_2}{2\pi v}\right) \quad \text{thus:}$$

$$\frac{ql}{m \times 2\pi v}[B_2 - B_1] = 1 \Rightarrow \frac{q}{m} = \frac{2\pi v}{l}[B_2 - B_1] \quad (1)$$

Now $qV = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2q}{m}V}$, using this in (1):

$$\frac{q}{m} = \frac{2\pi}{l}[B_2 - B_1] \cdot \sqrt{2V} \cdot \sqrt{q/m} \quad \Rightarrow \frac{q}{m} = \frac{8\pi^2 V (B_2 - B_1)^2}{l^2} \text{ ans.}$$

→ A solenoid is a current carrying metallic coil which creates a nearly uniform axial field inside it (provided coil is sufficiently long).



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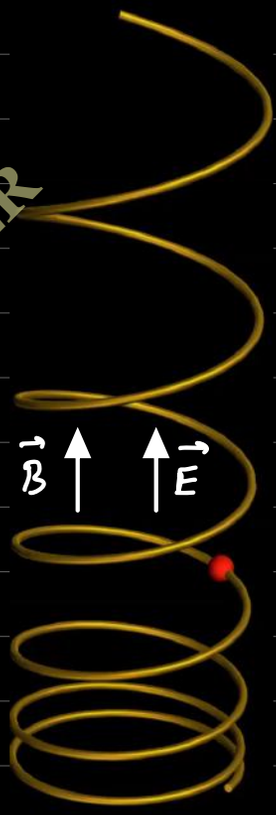
Lorentz Force:

The net electromagnetic force experienced by any charged particle is known as Lorentz Force. It is simply a superposition of Electric Force and Magnetic force. Thus, net Lorentz force is given as:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Motion In Electromagnetic Field ($\vec{E} \parallel \vec{B}$):

Recall that magnetic field exerts no force for velocity component parallel to $\vec{B}$, the magnetic force is always in a plane perpendicular to B, while the electric force is $q\vec{E}$ i.e along the electric field. Thus in case of uniform magnetic field, the projection of the trajectory in plane $\perp$ to B is circular with Larmor radius being given by $R = mv_\perp / qB$ (Note that since electric field provides no acceleration in plane $\perp$ to B thus $v_\perp$ remains constant in magnitude). The motion along the axis of the helix is uniformly accelerated i.e. $\vec{v}_\parallel = \vec{u}_\parallel + \frac{1}{2}\vec{a}t^2$ where $\vec{a}$ is simply $\vec{E}q/m$. This creates a trajectory in the shape of a helix of variable pitch. In case the electric force is against the initial $u_\parallel$, then after some time particle will have $v_\parallel$ against the initial $u_\parallel$ and a certain maximum/minimum coordinate will exist along $u_\parallel$ direction.



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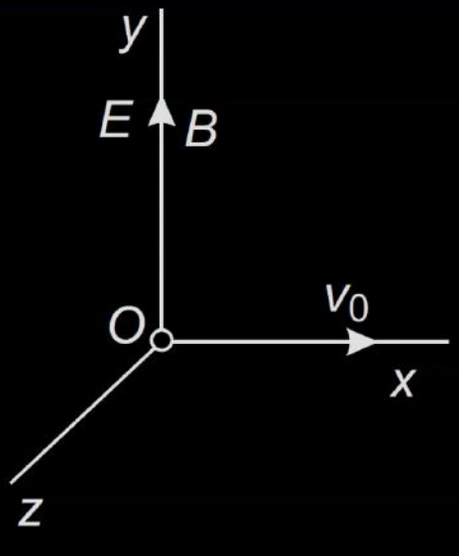
Example: Irodov: 3.387:

Uniform electric and magnetic fields with strength E and induction B respectively are directed along the y axis. A particle with specific charge q/m leaves the origin O in the direction of the x axis with an initial non-relativistic velocity v_0. Find: (a) the coordinate y_n of the particle when it crosses the y axis for the nth time; (b) the angle α between the particle's velocity vector and the y axis at that moment.



Solution: Time period $T = \frac{2\pi m}{qB}$. The particle will cross the y axis upon every integer multiple of T (Since initially it started from y axis's). Thus y coordinate on n^{th} crossing, $y_n = \frac{1}{2} a (nT)^2$

Thus $y_n = \frac{1}{2} \cdot (Eq/m) \cdot \left(\frac{2\pi m n}{qB}\right)^2 = \frac{2\pi^2 m E n^2}{qB^2}$ Ans.

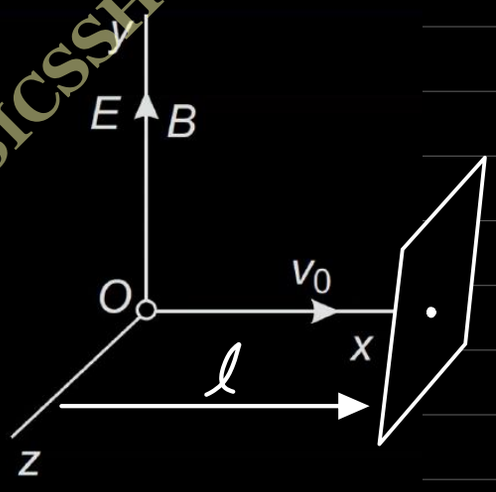
Now y component of velocity $v_{11} = u_{11} + a_{11} t = \frac{Eq}{m} \cdot n \times \frac{2\pi m}{qB}$

$\Rightarrow v_{11} = 2\pi n E/B$, Thus angle α with y axis is defined by:

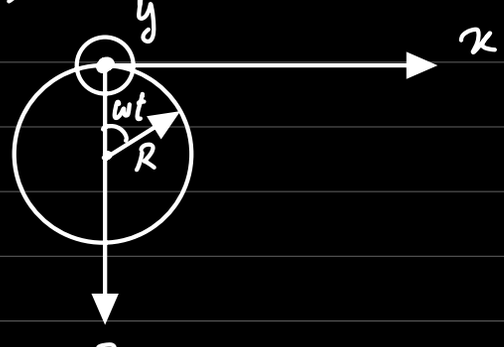
$\tan\alpha = v_\perp / v_{11} = Bv_0 / 2\pi n E$ Ans.

EXAMPLE: IRODOV: 3.388:

A narrow beam of identical ions with specific charge q/m, possessing different velocities, enters the region of space, where there are uniform parallel electric and magnetic fields with strength E and induction B, at the point O. The beam direction coincides with the x axis at the point O. A plane screen oriented at right angles to the x axis is located at a distance l from the point O. Find the equation of the trace that the ions leave on the screen. Demonstrate that at $z \ll l$ it is the equation of a parabola.



Solution: The screen is parallel to y-z plane so to find the trajectory, we need to find y and z coordinates at $x = l$. Now, $x = R\sin\omega t$, $z = R(1-\cos\omega t)$ and $y = \frac{1}{2} a t^2$, $R = v_0/\omega$.



Now $\because x = l$, $\sin\omega t = (\omega l / v_0)$

$\Rightarrow \omega t = \sin^{-1}(\omega l / v_0)$, Thus,

$$z = 2R\sin^2(\omega t/2) = \frac{v_0}{\omega} \cdot 2\sin(\omega t/2)\cos(\omega t/2) \cdot \frac{\sin(\omega t/2)}{\cos(\omega t/2)}$$

$$\Rightarrow z = \frac{v_0}{\omega}\sin(\omega t) \cdot \tan(\omega t/2) = l\tan(\omega t/2) = l\tan\left(\frac{qB}{2m}\left(\frac{2y}{Eq/m}\right)^{1/2}\right)$$

$$\Rightarrow z = l\tan\sqrt{\frac{qB^2 y}{2mE}}$$, Now if $z << l$,

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$$\tan\sqrt{\frac{qB^2y}{2mE}} \approx \sqrt{\frac{qB^2y}{2mE}} \quad (\because \tan\theta \approx \theta \text{ if } \tan\theta << 1)$$

Thus, $z = l\sqrt{\frac{qB^2y}{2mE}} \Rightarrow y = \frac{2mE}{qB^2l^2}z^2$, This is a parabola QED.

Uniform Motion In Electromagnetic Fields: For uniform motion, net force on the particle must be zero, thus:

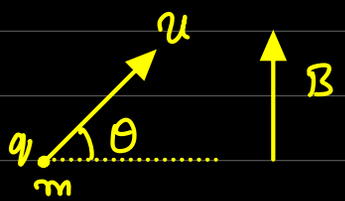
$q(\vec{E} + \vec{v}\times\vec{B}) = 0 \Rightarrow \vec{E} = -\vec{v}\times\vec{B}$. Whenever this happens, the particle will move with a constant velocity in the E.M. field.

→ In case of simultaneous presence of gravitational and Electromagnetic fields, gravitational field can be suitably accomodated in to an effective electric field. This is done as follows:

$$\vec{F} = q(\vec{E} + \vec{v}\times\vec{B}) + m\vec{g} \Rightarrow \vec{F} = q\left[\vec{E} + \frac{m\vec{g}}{q} + \vec{v}\times\vec{B}\right]$$

This can be seen as $\vec{F} = q(\vec{E}_{eff} + \vec{v}\times\vec{B})$ where $\vec{E}_{eff} = (\vec{E} + m\vec{g}/q)$ (Note that here sign of q is important).

Example: A particle of mass m and charge q is projected with speed u at an angle θ near the surface of earth, with a magnetic field pointing in upward direction. Find the permissible values of B so that particle finally lands at its launching point.



Solution: The motion of the particle is a helix with variable pitch. For particle to land at the starting point, the time of flight must be integer multiple of the Larmor time period.

Thus: $\frac{2\pi m}{qB} \times n = \frac{2u\sin\theta}{g} \Rightarrow B = \frac{\pi mgn}{qu\sin\theta}$ Ans.

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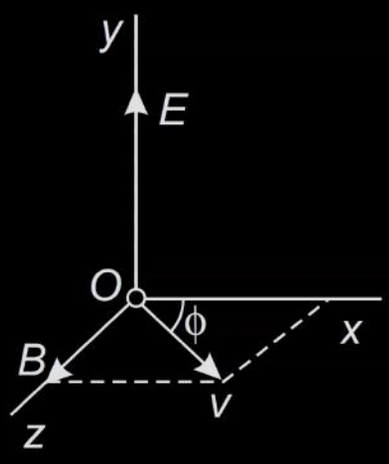
Example: Irodov: 3.390:

Non-relativistic protons move rectilinearly in the region of space where there are uniform mutually perpendicular electric and magnetic fields with $E = 4.0$ kV/m and $B = 50$ mT. The trajectory of the protons lies in the xz plane and forms an angle $\phi = 30°$ with the x axis. Find the pitch of the helical trajectory along which the protons will move after the electric field is switched off.



Solution: For proton to move uniformly, $(\vec{E} + \vec{v} \times \vec{B}) = 0$

Thus, $vB\cos\phi = E$ (1). Now pitch of the helix :

$\lambda = v \sin\phi \cdot \frac{2\pi m}{qB}$ (2) Now (2)/(1) yields: $\frac{\tan\phi}{B} \times \frac{2\pi mE}{qB} = \lambda$

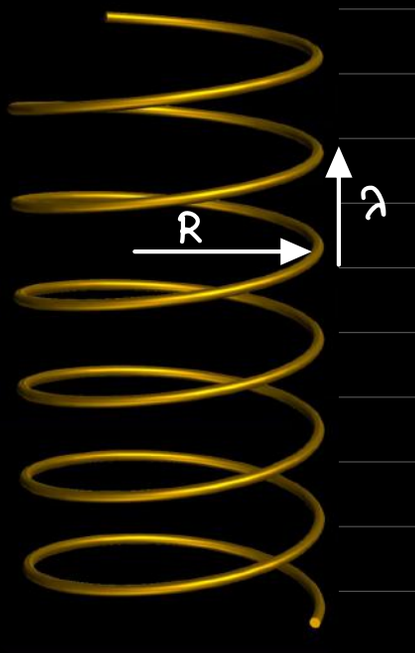
Thus required pitch, $\lambda = \frac{2\pi m \tan\phi E}{qB^2}$ Ans.

Substituting for the values, we get $\lambda = 6$ cm Ans.

Radius Of Curvature Of A Constant Pitch Helix :

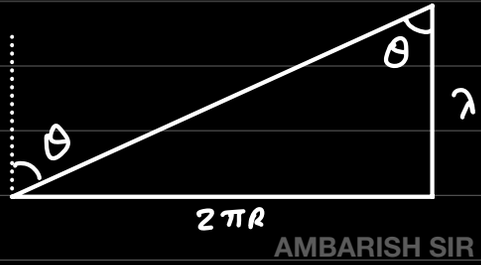
With constant magnetic field, the speed of particle remains constant. Thus the particle will have only centripetal acceleration, and radius of curvature of helix is simply v^2/a_c where v is speed of the particle and a_c is centripetal acceleration.

Now $a_c = \frac{v_1^2}{R}$, thus radius of curvature of helix, $r = \frac{v^2}{a_c} = R\left(\frac{v}{v_1}\right)^2$



Thus $r = R\,\mathrm{cosec}^2\theta$ where θ is angle between B and v.

$\boxed{r = R\,\mathrm{cosec}^2\theta}$ Now let us unwrap one turn, Consider the adjacent figure. $\tan\theta = \frac{2\pi R}{\lambda}$



$\Rightarrow \mathrm{cosec}\,\theta = \sqrt{4\pi^2R^2 + \lambda^2}\,/(2\pi R)$

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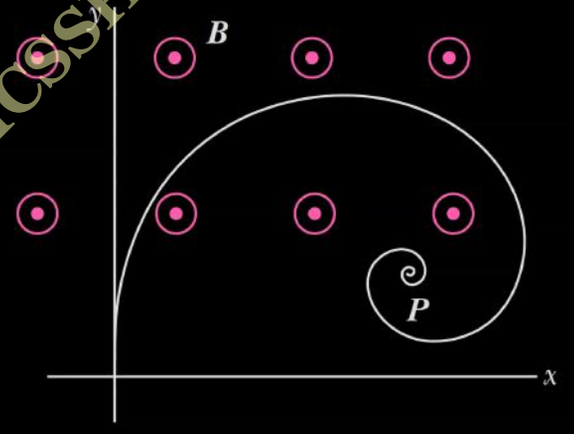
Thus radius of curvature of helix, $\eta = R\left[1+\frac{\lambda^2}{4\pi^2R^2}\right]$ Ans.

Projection In Simultaneous Magnetic And Frictional Forces:

Whenever a particle is projected is simultaneous presence of magnetic and viscous or kinetic frictional forces, the tangential acceleration is only provided by the frictional or viscous forces and the centripetal acceleration is provided only by magnetic forces. It must also be kept in mind that in such situations, the loss in mechanical energy occurs only due to frictional or viscous forces.

Example:

Let the first and second quadrant contain a constant magnetic field $\vec{B} = B\hat{k}$ ($\vec{B}$ points out of the page). A particle with charge q and mass m travels along the y axis and enters the first quadrant with speed v_0. Assume that the particle is subject to a friction force, $\vec{F}_f$, proportional to its velocity, i.e., $\vec{F}_f = -\alpha\vec{v}$. (You may ignore gravity.) Assume this friction force is large enough so that the particle will remain in the first quadrant at all times. The particle will then spiral in toward the point P. (a)What are the coordinates of the point P ? (b) After how much minimum time does the velocity vector reverse its direction? (c) what is the distance travelled by the particle as a function of time? (d) what is the velocity vector as a function of time?



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Solution: Here for solving part (a) we can form a differential equation to solve for $\vec{v}(t)$ and then integrate it to find total displacement, howeveven we shall see that there is a shorten method based on impulse momentum theorem. Here: $\vec{F} = -\alpha\vec{v} + q\vec{v}\times\vec{B}$

Now integrating $\vec{F}$ wrt t, $\vec{J} = \int\vec{F}dt = -\alpha\vec{s} + q\vec{s}\times\vec{B}$.

Now let finally $\vec{s} = a\hat{i} + b\hat{j} \Rightarrow \vec{J} = -\alpha(a\hat{i}+b\hat{j}) + q(a\hat{i}+b\hat{j})\times B\hat{k}$

$\Rightarrow \vec{J} = \hat{i}[-\alpha a + qBb] + \hat{j}[-\alpha b - aqB] = \Delta\vec{p} = -mv_0\hat{j}$

By Comparison:

$-\alpha a + qBb = 0$ (1) and $-\alpha b - aqB = -mv_0$ (2)

Solving: $a = \frac{qBm}{q^2B^2+\alpha^2}v_0$ and $b = \frac{m\alpha v_0}{q^2B^2+\alpha^2}$ Ans

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(b) Now $qvB = mv\omega$ ($\because F_c = ma_c$ and $a_c = v\omega$)

$\Rightarrow \omega = qB/m$ (Note that ω is rate of rotation of $\vec{v}$ vector).

Thus rate of rotation of velocity vector is independent of time).

Now for velocity vector to reverse, it must turn through an angle π.

thus required time for reversal of direction, $T_h = \pi/\omega = \frac{\pi m}{qB}$ Ans.

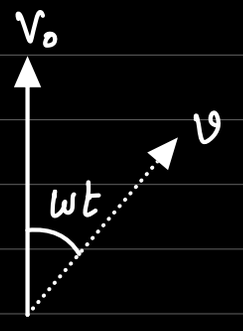
(c) Tangential force $F_t = -\alpha v = m\,dv/dt$, thus $dv/v = -(\alpha/m)\,dt$

$$\Rightarrow \int_{v_0}^{v} \frac{dv}{v} = -\frac{\alpha}{m}\int_0^t dt \quad \Rightarrow \quad v = v_0 e^{-\alpha t/m} = dl/dt$$

$$\Rightarrow l = v_0\int_0^t e^{-\alpha t/m}\,dt = -\frac{mv_0}{\alpha}\left[e^{-\alpha t/m}\right]_0^t = \frac{mv_0}{\alpha}\left[1 - e^{-\alpha t/m}\right]$$

Thus $l = \frac{mv_0}{\alpha}\left[1 - e^{-\alpha t/m}\right]$ Ans.

(d) As seen in previous parts, $v = v_0 e^{-\alpha t/m}$ and angle through which the vector turns is $\theta = \omega t$ where $\omega = qB/m$. Thus velocity as a function of time is given as:



$$v = v_0 e^{-\alpha t/m}\left[\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j}\right]$$ Ans.

Field Transformation Equations (Non-Relativistic):

Let $\vec{E}$ and $\vec{B}$ be the Electric and Magnetic fields respectively in a frame K and let a frame K' move with a velocity $\vec{v}_0$ relative to K, then, the Electric and magnetic fields as seen from K' are given by:

$$\boxed{\vec{E'} = \vec{E} + (\vec{v}_0 \times \vec{B})}$$

$$\boxed{\vec{B'} = \vec{B} - \frac{\vec{v}_0 \times \vec{E}}{c^2}}$$

→ Since $\vec{v}_0 \times \vec{B}$ and $\vec{v}_0 \times \vec{E}$ terms are $\perp$ to $\vec{v}_0$ thus it means that field components parallel to velocity remain unchanged i.e

$E'_{\parallel} = E_{\parallel}$ and $B'_{\parallel} = B_{\parallel}$

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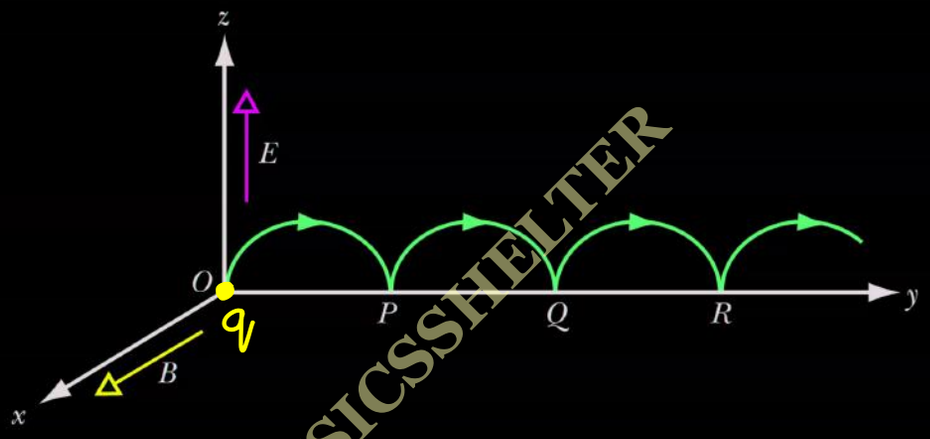
→ For simplifying the analysis of problems involving projection of a charged particle in an electromagnetic field, it is often convenient to find a frame in which $\vec{E}' = 0$. In such a frame, we only have to deal with magnetic forces.

→ In most problems, $\vec{v}_0 \times \vec{E} << c^2$ thus for calculations we may assume $\vec{B}' \approx \vec{B}$.

EXAMPLE: A particle is released from origin as shown in the figure. There exist uniform electric and magnetic fields in the region such that, $\vec{E} = E\hat{k}$ and $B = B\hat{i}$



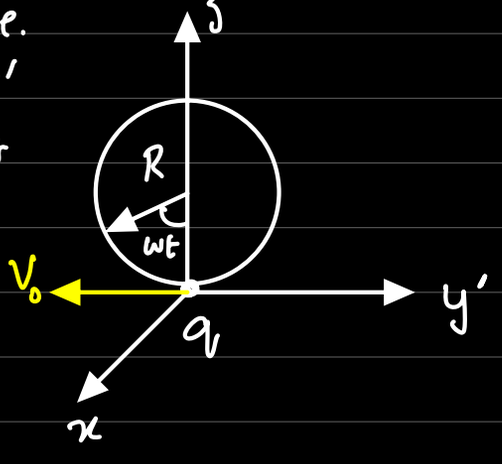
Find the y and z coordinates of the particle as a function of time. (Use field transformation approach).

Solution: Imagine a frame moving with a speed E/B along the direction of $\vec{E} \times \vec{B}$ i.e $\hat{j}$. We shall show that this frame will make the electric field zero. From the previous section:

$\vec{E}' = \vec{E} + (\vec{v}_0 \times \vec{B})$, thus: $\vec{E}' = E\hat{k} + \left(\frac{E}{B}\hat{j}\right) \times (B\hat{i}) = 0$

Now we shall find coordinates of particle in K' frame and then add $v_0 t\hat{j}$ to get coordinates in the original frame.

Consider the adjacent figure drawn from K' frame. (Note that x and z coordinates are same in K' and hence the labels, though y' coordinate is different due to $\vec{v}_0$ being along y axis).



Thus, $y' = -R\sin(\omega t)$, $z = R(1-\cos\omega t)$

where $R = \frac{mv_0}{qB}$, $\omega = qB/m$.

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Now $y = v_0 t + y' = \frac{E}{B} t - \frac{m(E/B)}{qB} \sin\left(\frac{qB}{m} t\right)$

$\Rightarrow y = \frac{E}{B}\left[t - \frac{1}{\omega}\sin(\omega t)\right] = \frac{E}{B\omega}\left[\omega t - \sin(\omega t)\right]$ Ans.

$z = \frac{m(E/B)}{qB}\left[1 - \cos\omega t\right] = \frac{E}{\omega B}\left[1 - \cos\omega t\right]$ Ans

Note that from x' frame, the particle is executing a uniform circular motion with a constant speed v_0 while from x frame the centre of this circle is advancing with a speed v_0. This situation is just like that of a particle on periphery of a rolling cylinder, i.e a cycloid.

Following Points Must Be Memorized For Particle Released From Rest With E ⊥ B:

1. The path is a cycloid with the rolling cylinder with following parameters:

   Guiding centre speed: $\boxed{v_0 = E/B}$

   Angular speed of rolling: $\boxed{\omega = qB/m}$

   Radius of cylinder: $\boxed{R = v_0/\omega = mE/qB^2}$

2. Plane of motion is perpendicular to $\vec{B}$

3. Direction of drift of the guiding centre is $\vec{E} \times \vec{B}$

4. Sense of rotation of cylinder is given by right hand thumb rule (i.e for electron, point the thumb along $\vec{B}$ and curl of fingers gives sense of rotation of the cylinder)

5. Speed of particle at any coordinate can be simply found using work energy theorem, where work done only depends on coordinate along Electric field because magnetic field does no work. Thus $qE \times s_E = \frac{1}{2} mv^2$ where s_E is component of displacement along electric field.

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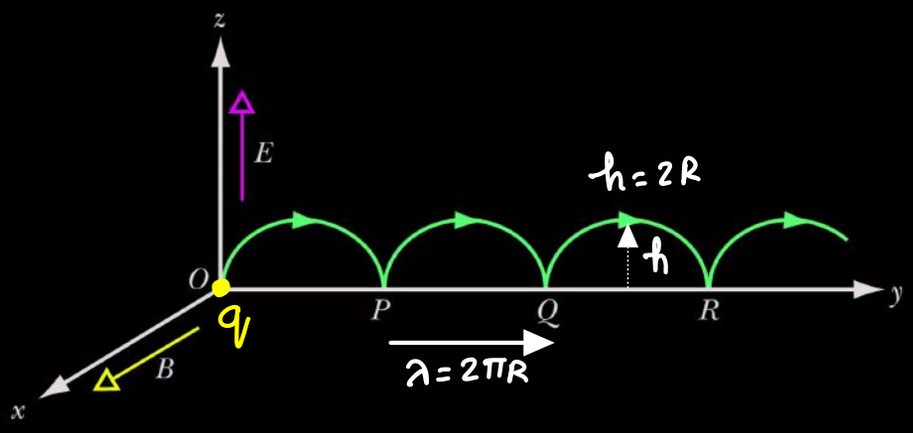
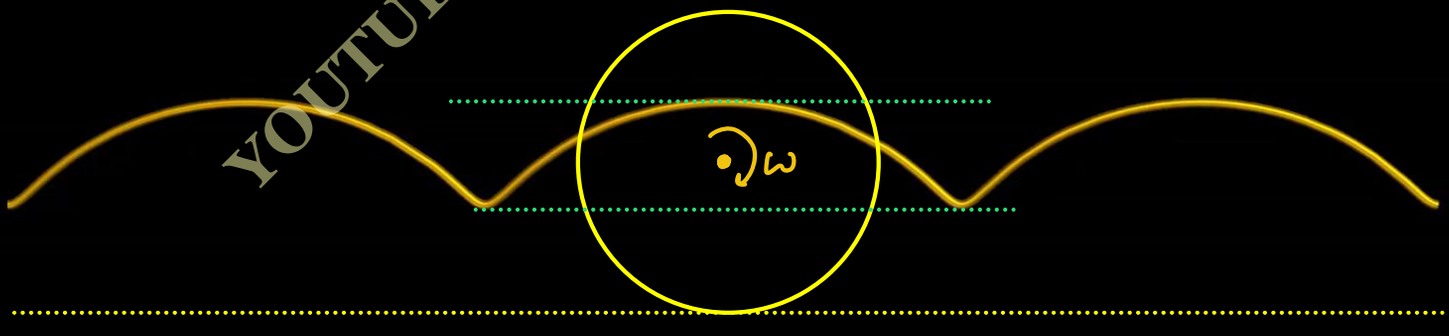
6. The particle repeatedly touches the rolling line at a periodic distance interval $\lambda = 2\pi R$ and maximum distance from rolling line being $h = 2R$.



7. If instead of being released from rest, the particle is projected with some speed in a plane perpendicular to B, the path may be either a cycloid or an inferior trochoid or a superior trochoid based on following cases:

→ If projection speed in the guiding centre frame is equal to ωR, the path is a cycloid.

→ If projection speed in guiding centre frame is less than ωR, the path is inferior trochoid:



→ If projection speed in guiding centre frame is more than ωR then the path is a superior trochoid.

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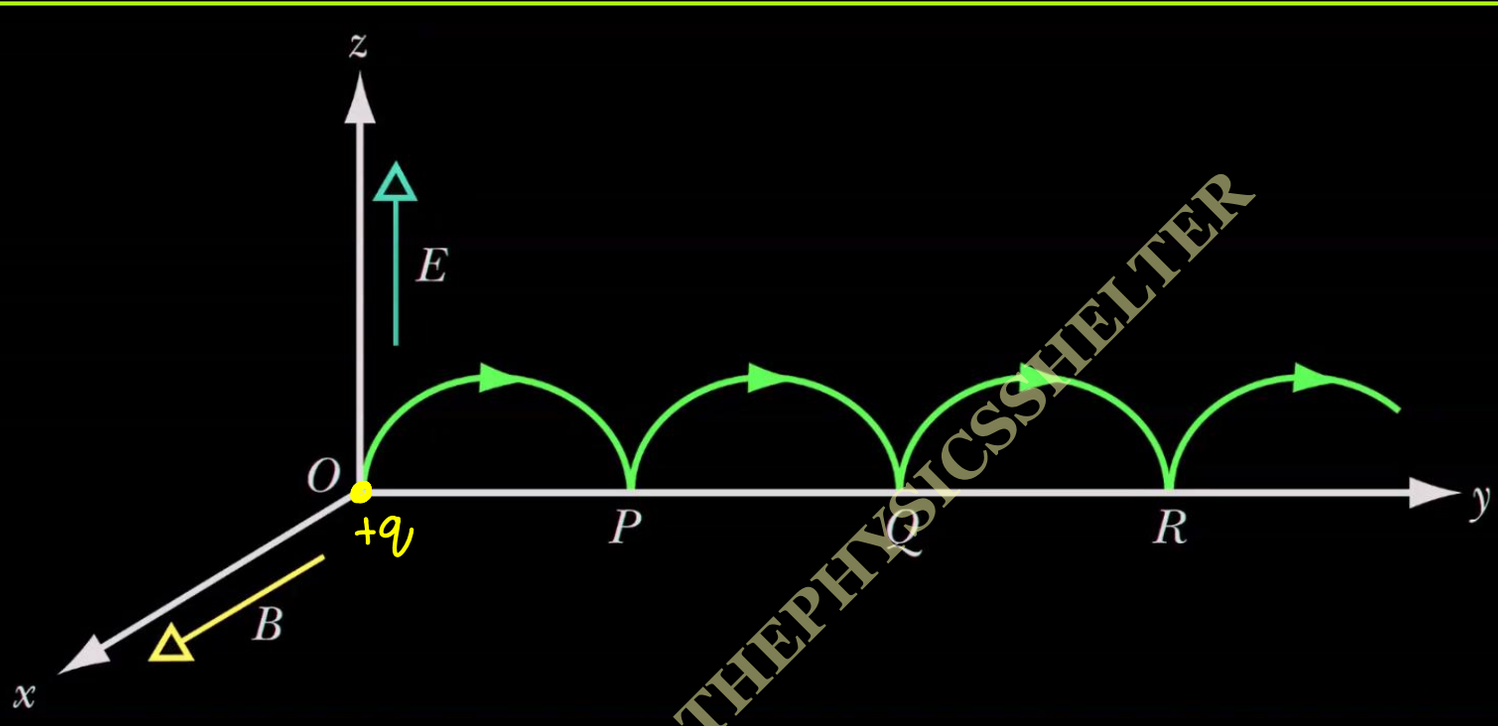
Motion Of Charged Particle In Uniform E & B (E ⊥ B) (Without Field Transform):



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Consider a charged particle released in Electromagnetic field with $\vec{E} = E\hat{k}$ and $\vec{B} = B\hat{i}$. Now since magnetic force and electric force are both confined to the yz plane, let velocity of particle at a general time be:

$\vec{v} = v_y\hat{j} + v_z\hat{k}$, Now Lorentz Force, $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ i.e:

$$\vec{F} = q\left(E\hat{k} + (v_y\hat{j} + v_z\hat{k}) \times B\hat{i}\right), \quad \text{Thus:}$$

$$F_z = q(E - v_y B) = m\,dv_z/dt \quad (1) \quad \text{and}$$

$$F_y = q v_z B = m\,dv_y/dt \quad (2)$$

Now differentiating (1) wrt time: $-qB(dv_y/dt) = m\,d^2v_z/dt^2$ (3)

Now substituting for dv_y/dt from (3) in (2):

$$q V_3 B = m \times \left(\frac{-m}{qB}\right) \frac{d^2 V_3}{dt^2} \Rightarrow \frac{d^2 V_3}{dt^2} = -\omega^2 V_3 \quad (4)$$

where $\omega = qB/m$.

Now equation (4) is of the same form as SHM differential equation, thus:

$V_3 = V \sin(\omega t + \phi)$ (5) and V and ϕ must be evaluated to satisfy initial conditions i.e, @ $t=0$, $V_3 = 0$ and $dV_3/dt = Eq/m$,

Thus putting these conditions in (5):

$0 = V \sin(\phi)$ and $\frac{Eq}{m} = V\omega \cos\phi \Rightarrow \phi = 0, \; V = \frac{Eq}{m\omega}$

i.e $V = \frac{Eq}{m \times (qB/m)} = E/B$. Thus: $V_3 = \frac{E}{B} \sin(\omega t)$ Ans.

Now from (2), $\frac{dV_y}{dt} = \omega V_3 = \omega \frac{E}{B} \sin(\omega t)$

$$\Rightarrow V_y = \frac{E\omega}{B} \int_0^t \sin\omega t = \frac{E}{B} [1 - \cos\omega t] \text{ Ans.}$$

Now $y = \int_0^t V_3 \, dt = \frac{E}{B} \int_0^t \sin(\omega t) dt = \frac{E}{B\omega} [1 - \cos\omega t]$ Ans.

(This is the same result that we obtained earlier by field transformation)

Now $y = \int V_y dt = \frac{E}{B} \int_0^t 1 - \cos\omega t \, dt = \frac{E}{B} \left[t - \frac{\sin\omega t}{\omega}\right]_0^t$

$\Rightarrow y = \frac{E}{B\omega} [\omega t - \sin\omega t)$ Ans. (This is also the same result as before).

Projection In Crossed Gravitational And Magnetic Fields ($\vec{g} \perp \vec{B}$):

From dynamics point of view, just like a constant electric field, a constant gravitational field applies a constant force and can be modelled as an equivalent Electric field $\vec{E}$ such that:

$q\vec{E} = m\vec{g}$ or $\vec{E} = \frac{m\vec{g}}{q} = \vec{g}/\alpha$ where α is specific charge.

Now the standard theory for trochoidal motion can be applied to the questions involving crossed $\vec{g}$ and $\vec{B}$ fields.

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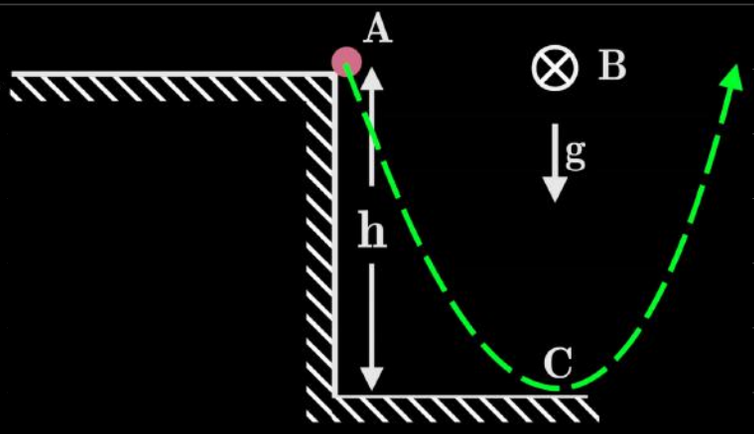
EXAMPLE :

A proton of mass m and charge q is released from rest from the top A of the building of height h in the presence of uniform gravitational and magnetic fields as shown. The height of the building is chosen such that the proton during its flight just touches the ground for the first time at C. Neglect air resistance and mark the correct statement(s):



(A) Distance travelled by the particle from A to C is 2h

(B) Horizontal separation between the points A and C is 2h

(C) Velocity of the particle at C is $2\sqrt{gh}$

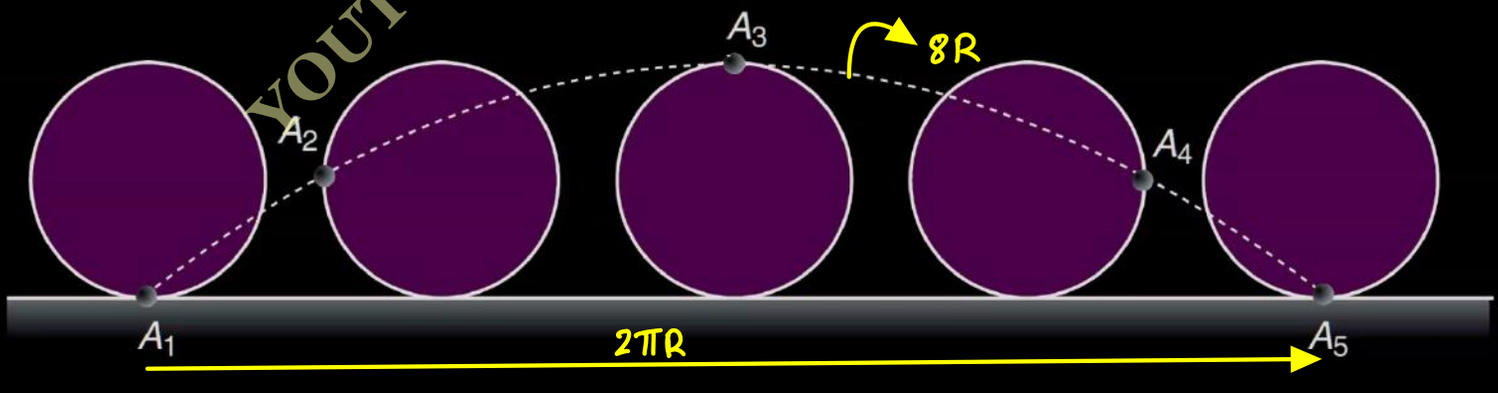
(D) Time of fight of the particle from A to C is $\frac{\pi m}{2qB}$

Solution: As discussed in the previous section, this problem is reduced to the standard trochoidal motion problem. Now the parameters of motion:

$$\omega = qB/m, \quad R = E/B\omega = g/\omega B\omega = mg/qB\omega.$$

Now recall from rotational mechanics that length of a cycloid during one cycle is 8R. In our problem, AC is half of this length, thus distance AC = 4R, and total depth = 2R = h. Thus distance AC = 2h thus option (a) is correct.

Now Horizontal separation from A to C = πR (Recall from rolling cylinder motion in rotational mechanics). Thus B option is incorrect.



Now velocity at C can be found using energy conservation, thus $v_c = \sqrt{2gh}$, thus C option is incorrect.

Now time of flight from A to C is half cycle of rolling cylinder i.e $\pi/\omega = \pi/(qB/m) = \pi m/qB$. Thus D is also wrong.

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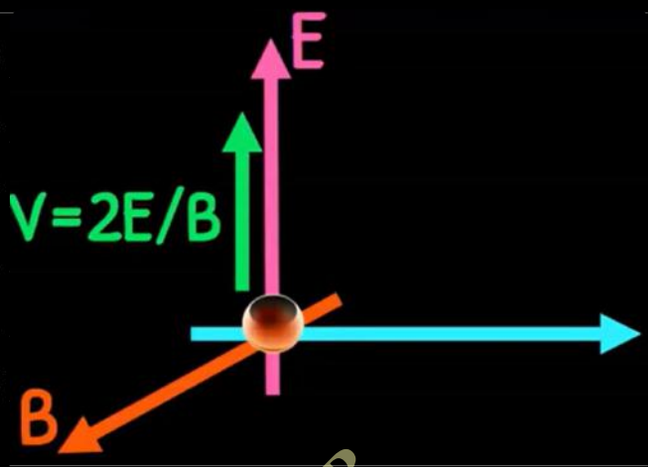
Thus only correct option is (a).

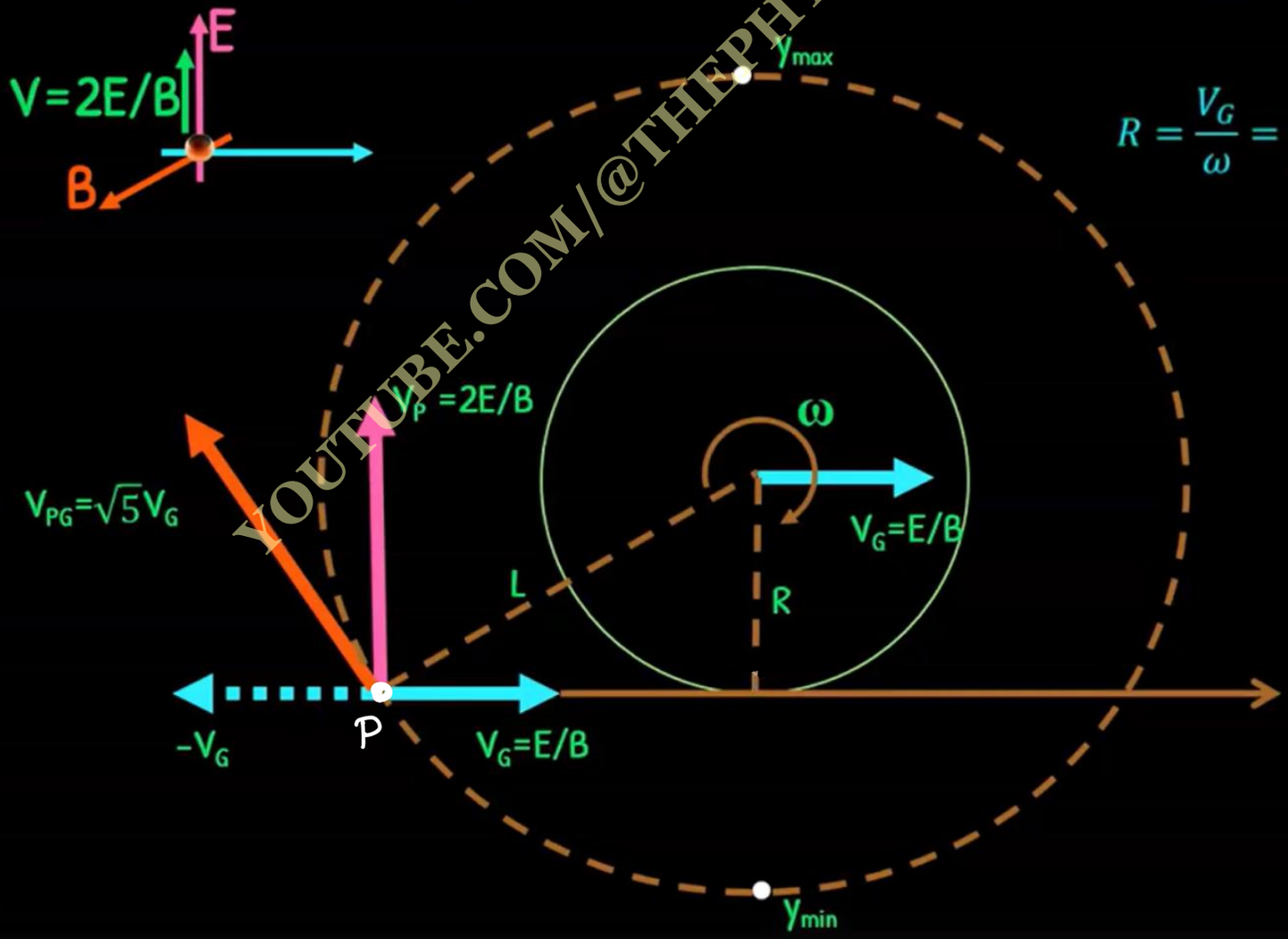
EXAMPLE:

A charged particle having charge q and mass m is projected from origin along y axis with a velocity $v=\frac{2E}{B}\hat{j}$ in simultaneous presence of a uniform Electric field $E\hat{j}$ and a uniform Magnetic field $B\hat{k}$. Calculate the maximum and minimum y coordinates of the particle during the subsequent motion.



Solution: Velocity of Guiding centre: $\vec{v}_0 = \frac{E}{B}\hat{i}$ (i.e along $\vec{E}\times\vec{B}$).

Now velocity relative to guiding centre: $(2E/B)\hat{j} - (E/B)\hat{i}$, its magnitude, $v = \sqrt{5}E/B$ which is greater than V_0, thus the path is a superior trochoid.



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Now Consider the above figure. The guiding centre is located by drawing $\perp$ to V_{PA} of length $L = V_{PA}/\omega$. Now as clear from figure, $y_{min} = -(L-R)$ and $y_{max} = R+L$. Now let us find L and R. $\omega = qB/m$, $R = V_0/\omega = E/B\omega = Em/qB^2$

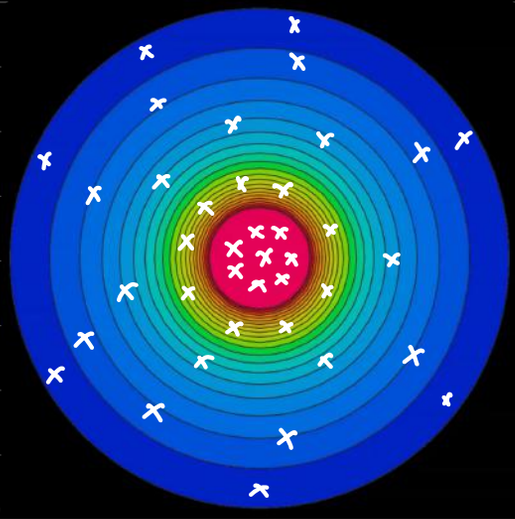
$L = V_{PA}/\omega = \sqrt{5}\, Em/qB^2$. Thus $y_{max} = (\sqrt{5}+1)\, mE/qB^2$ Ans

and $y_{min} = -(\sqrt{5}-1)\, mE/qB^2$ Ans.

→ Initial position of guiding centre is on $\perp$ to V_{PA} at a distance of V_{PA}/ω. Direction of Rolling is $\vec{E}\times\vec{B}$, and radius of rolling cylinder is V_0/ω and sense of rolling is given by the thumb rule (Right hand for electron and left hand for +ve charge).

MAGNETIC FLUX: Just as we defined electric flux through any patch as $d\phi = \vec{E}\cdot d\vec{S}$ and $\phi = \int \vec{E}\cdot d\vec{S}$ similarly we can also define magnetic flux as $\phi_B = \int \vec{B}\cdot d\vec{S}$. Just like electric flux, there are also some physical significances associated with magnetic flux about which we shall study in due course of time. In a short while we will be seeing the relationship between change in angular momentum and magnetic flux for cylindrically symmetric axial magnetic fields.

EXAMPLE: An axial magnetic field exists in a cylindrical region of radius R such that field varies as $B = \frac{B_0 R}{r}$ where r is the distance from the centreline of the cylinder.



(a) Find magnetic flux through the cross section disc of radius R.

(b) Find ratio of magnetic field at periphery to the average field over the cross section (Average field here means ratio of flux to Area of cross section).

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Solution: Consider an annulus of radius r and radial thickness dr.

Then, $d\phi_B = B \times 2\pi r\,dr = \frac{B_0 R}{r} \times 2\pi r\,dr = 2\pi B_0 R\,dr$

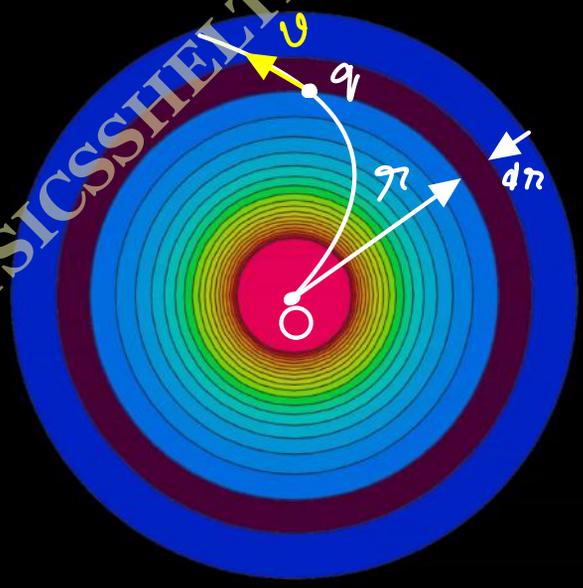
$\Rightarrow \phi_B = 2\pi B_0 R \int_0^R dr = 2\pi B_0 R^2$ Ans. Now average field :

$B_{avg} = \phi_B / A = 2\pi B_0 R^2 / \pi R^2 = 2B_0$, Field at periphery (i.e $r = R$),

$B_p = B_0$ thus required ratio, $B_p / B_{avg} = 1/2$ Ans.

ANGULAR MOMENTUM CHANGE WHILE CROSSING AN ANNULUS OF AXIAL FIELD:

Consider a charged particle crossing an annular region as shown in adjacent figure. Magnetic torque about point O experienced by the particle:



$$\vec{\tau} = \vec{r} \times q(\vec{v} \times \vec{B})$$

$$\Rightarrow \vec{\tau} = q[\vec{v}(\vec{r}\cdot\vec{B}) - \vec{B}(\vec{r}\cdot\vec{v})]$$

$$\Rightarrow \vec{\tau} = q\left(-\vec{B}\, r \frac{dr}{dt}\right) \Rightarrow \vec{\tau}\,dt = -q\vec{B}\, r\,dr$$

Thus $d\vec{L} = -q\vec{B}\, r\,dr$, Now flux through the annulus, $d\phi = B \times 2\pi r\,dr$

$\Rightarrow B r\,dr = \frac{d\phi}{2\pi}$, thus, $d\vec{L} = -\frac{q\,d\phi}{2\pi}\hat{B}$

or $\boxed{|dL| = \frac{q\,d\phi}{2\pi}}$

→ This result can be conveniently used to find the angle of velocity vector with the radius vector at any radius.

EXAMPLE: A charged particle is launched radially in a cylindrically symmetric axial field $B = B_0 R/r$. The mass of particle is m, its charge q and projection speed v. At what angle with the radial direction does the particle emerge?

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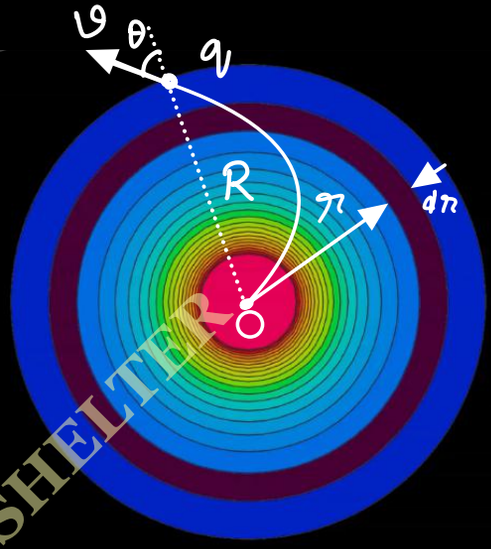
Solution: Recall from previous section, $d\vec{L} = -\frac{q\,d\phi}{2\pi}\hat{B}$, thus total change in angular momentum, $\Delta\vec{L} = -\frac{q\hat{B}}{2\pi}.(\Delta\phi)$, Now $\Delta\phi$ as calculated in a previous example, $\Delta\phi = 2\pi B_0 R^2$

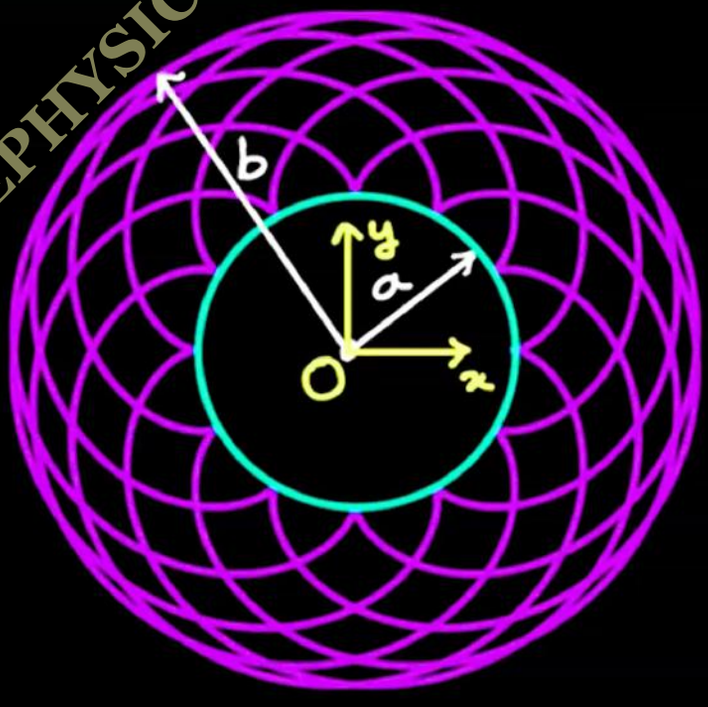


Now initial angular momentum was zero and final angular momentum is $mvR\sin\theta$, thus:

$$mvR\sin\theta = \frac{2\pi\, q B_0 R^2}{2\pi} \Rightarrow \theta = \sin^{-1}\left(\frac{B_0 Rq}{mv}\right) \text{ Ans.}$$

EXAMPLE (ORIGINAL):

A charged particle having charge q and mass m released from rest in an electromagnetic field traces a path as shown in the adjacent figure (magenta color). The region has a cylindrically symmetric radial electric field and a cylindrically symmetric uniform axial magnetic field B. The nearest and farthest distances of the particle from the center of the figure are a and b respectively. The speed of the particle at the farthest point is v. Find the magnitude of the magnetic field. (There are no physical obstacles, the circle is just drawn to show the boundary of nearest point.



Solution: Since the electric field is radial, it contributes nothing to change in angular momentum about centre of the field. Also recall that the change in angular momentum in this situation is related to magnetic flux in the region. At the inner radius, we see a sharp turn in the path, which means that velocity here must be zero (otherwise there will be infinite accelerations).

Now $\Delta L = \frac{q\Delta\phi}{2\pi}$, thus: $mvb = \frac{q \times \pi(b^2-a^2)B}{2\pi}$

$$\Rightarrow B = 2mvb/q(b^2-a^2) \text{ Ans.}$$

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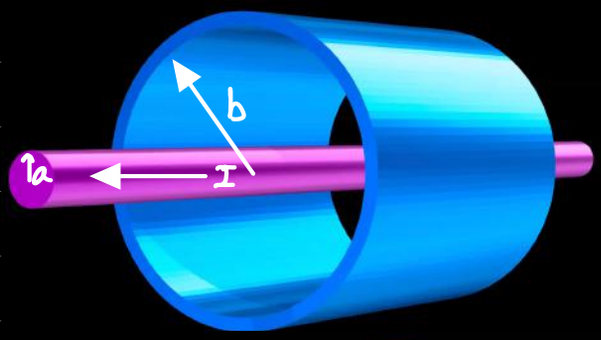
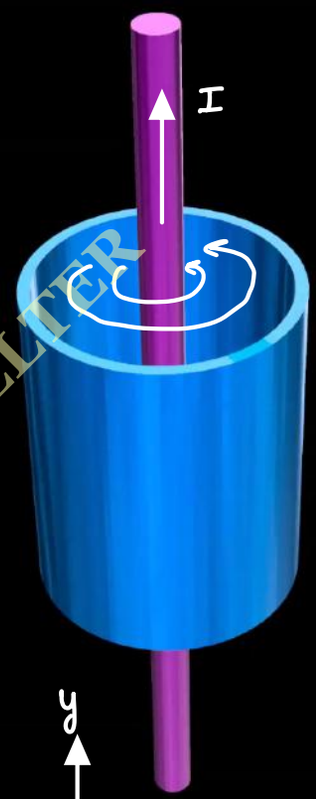
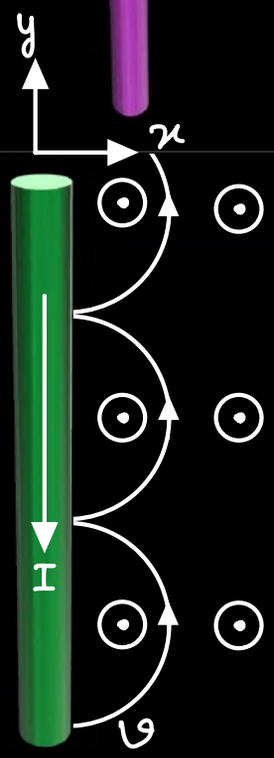
EXAMPLE: An electron leaves surface of a current carrying wire with negligible velocity due to thermionic emission. The ejection velocity v is perpendicular to the surface of inner electrode. The radius of current carrying wire is a and there is a concentric cylindrical structure around the wire. If the current in the wire is I, find the maximum speed with which the electrons must be ejected so that the electrons do not hit the outer cyclinder.





Given that magnetic field due to the current carrying wire is $\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{\theta}$

where r is the radial distance from the axis. The radius of current carrying wire is b.



Solution: Let the velocity of electron at a general point be $v_x\hat{i} + v_y\hat{j}$, Now Lorentz force:

$$\vec{F} = -e\vec{v}\times\vec{B} = -e(v_x\hat{i} + v_y\hat{j})\times\left(\frac{\mu_0 I}{2\pi x}\hat{k}\right)$$

$$\Rightarrow \vec{F} = \frac{\mu_0 I e}{2\pi x}\left[v_x\hat{j} - v_y\hat{i}\right]$$

Thus $a_x = F_x/m = -\frac{\mu_0 I e v_y}{2\pi m x}$

Thus $v_x\frac{dv_x}{dx} = -\frac{\mu_0 I e}{2\pi m x}\sqrt{v^2 - v_x^2}$ (1)

($\because v_x^2 + v_y^2 = v^2$ which remains constant since magnetic force does no work).

Now separating the variables and integrating:

$$\int_v^0 \frac{v_x\,dv_x}{\sqrt{v^2 - v_x^2}} = \int_a^b \frac{-\mu_0 I e}{2\pi m x}\,dx, \quad (\because \text{at max } x \text{ coordinate } v_x = 0 \text{ and } x_{max} = b)$$

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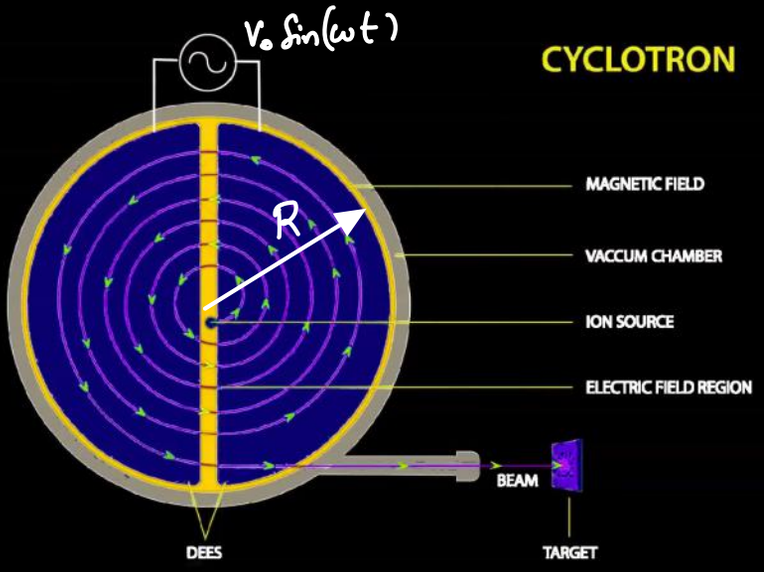
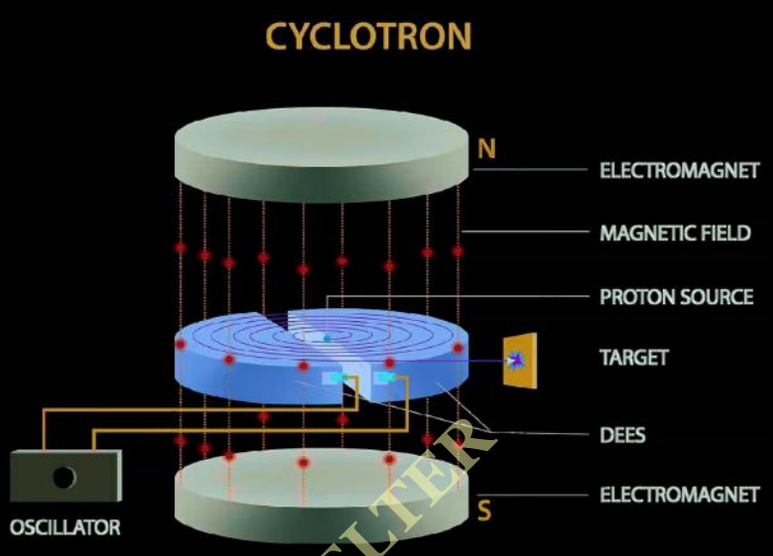
Thus $-v = -\frac{\mu_0 I e}{2\pi m} \ln\left(\frac{b}{a}\right) \Rightarrow v = \frac{\mu_0 I e}{2\pi m} \ln(b/a)$ Ans.

CYCLOTRON:





A cyclotron is a machine which is used to accelerate ions and charged particles. It consists of two D shaped structures. There is an alternating voltage applied between the gap between the Dees. There is a magnetic field perpendicular to the plane of the Dees. The ions begin to move along a circular path with angular frequency, $\omega = qB/m$. Due to accelerating voltage, as the particle crosses from one Dee to the other, particle gains kinetic energy and hence the radius of the path increases. By the time particle covers another semi-circle, the polarity of accelerating voltage reverses and once again particle gains kinetic energy and the radius again increases. The cycle keeps on repeating and the radius keeps increasing as the particle keeps gaining speed. The frequency of alternating voltage exactly matches the angular frequency of rotation of the particle (∵ polarity has to reverse every half circle). If the radius of Dee is R and initial speed is negligible then, $R = mv/qB$ or $\boxed{v = qBR/m}$, number of passes across the Dee Gaps can be calculated using work energy theorem. Thus $nqV_0 = \frac{1}{2}mv^2$

$\Rightarrow n = \frac{mv^2}{2qV_0}$ where v is final speed and V_0 is Voltage amplitude.

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Example: Irodov 3.397:

Protons are accelerated in a cyclotron so that the maximum curvature radius of their trajectory is equal to r=50 cm. Find:

(a) the kinetic energy of the protons when the acceleration is completed if the magnetic induction in the cyclotron is B=1.0 T;

(b) the minimum frequency of the cyclotron's oscillator at which the kinetic energy of the protons amounts to E=20 MeV by the end of acceleration.

(1 eV is the kinetic energy that an electron gains being accelerated by a potential difference of 1Volt, thus $1eV=1.6\times10^{-19}$ Joules)

Solution: (a) $r = mv/qB = \sqrt{2Em}/(qB) \Rightarrow E = q^2B^2r^2/2m$ Ans.

(Substituting the values, $E = 12$ MeV.

(b) For achieving a higher kinetic energy, a higher value of magnetic field will be required. Now, from part (a):

$$E = \left(\frac{qB}{m}\right)^2 \cdot \frac{1}{2}mr^2 = \frac{1}{2}mr^2\omega^2 \Rightarrow \omega = \frac{1}{r}\sqrt{\frac{2E}{m}}$$

Now frequency $\nu = \omega/2\pi = \frac{1}{\pi r}\sqrt{\frac{E}{2m}}$ Ans.

Putting the values, $\nu = 20$ MHz. (Note that any odd multiple of this frequency will also do the Job, ∵ odd multiple means that A.C. might complete either 1/2 cycle or 3/2 cycle of 5/2 cycle etc and each of them would lead to a reversal of polarity every time the charge completes a semi-circle).

Example: Irodov 3.396:

The cyclotron's oscillator frequency is equal to $\nu = 10$ MHz. Find the effective accelerating voltage applied across the Dees of that cyclotron if the distance between the neighbouring trajectories of protons is not less than $\Delta r = 1.0$ cm, with the trajectory radius being equal to $r = 0.5$ m.

Solution: $E = \frac{1}{2}mr^2\omega^2 = 2\pi^2 m r^2 \nu^2$

$\Rightarrow \Delta E = 2qV = 2\pi^2\nu^2[2r\Delta r]$ (∵ In the limiting case we choose exact Δr, and there are two passes/cycle).

$\Rightarrow V = \frac{2\pi^2\nu^2 r\Delta r}{q}$ Ans.

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Putting the values, we get $V = 0.1\,MV$.

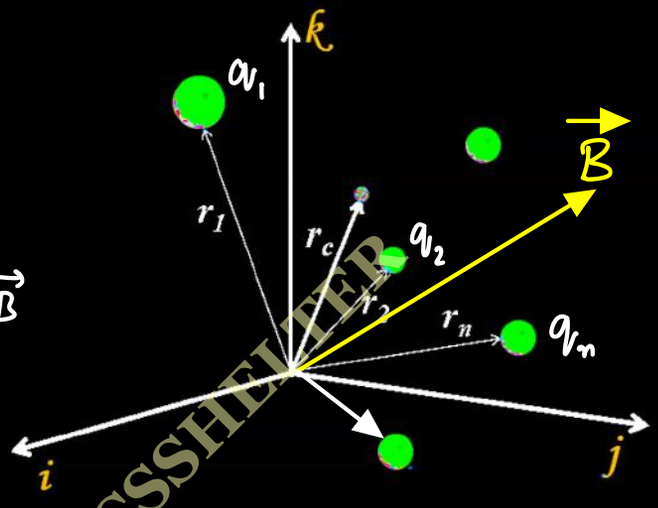
→ Note that $\Delta r \propto 1/p$ thus as the orbit size increases the circles come nearer and nearer.

Magnetic Force On A System Of Particles In Uniform Magnetic Field:

Consider a system of several particles moving with certain velocities in a uniform magnetic field $\vec{B}$. Now, net force on the system:

$$\vec{F} = \sum q_i \times (\vec{v}_i \times \vec{B}) = \left(\sum q_i \vec{v}_i\right) \times \vec{B}$$

$$\Rightarrow \vec{F} = Q\left(\frac{\sum q_i \vec{v}_i}{Q}\right) \times \vec{B}$$



where Q is the net charge of the system. Now $\left(\sum q_i \vec{v}_i\right)/Q = \vec{v}_{cm}$ i.e velocity of centre of charge. Thus net force on the system:

$$\boxed{\vec{F} = Q\,\vec{v}_{cm} \times \vec{B}}$$

→ This formula may be used for finding forces on charged moving bodies.

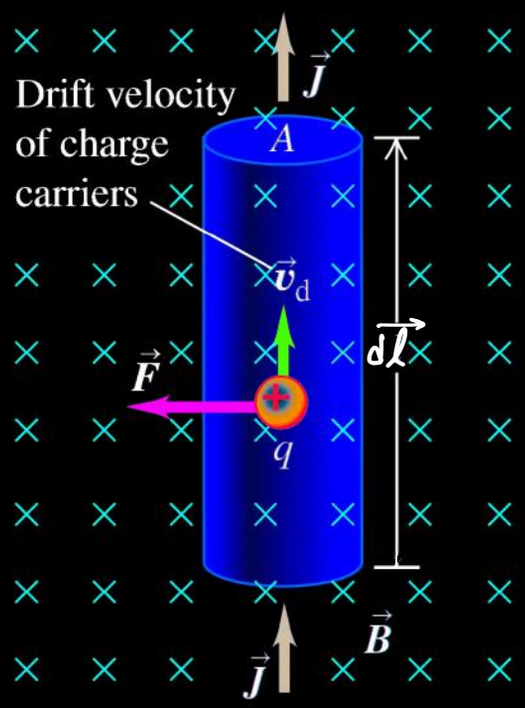
Magnetic Force On A Current Carrying Wire:

Recall from previous section that net force on collection of a system of particles is given as $\vec{F} = Q\,\vec{v}_{cm} \times \vec{B}$, Now consider a small element of current carrying wire having a length $d\vec{\ell}$. If n is density of free electrons, A the cross section area and drift velocity $\vec{v}_d$ then the force on this wire element is given as:

$$d\vec{F} = (nA\,d\ell)\,e\,\vec{v}_d \times \vec{B},$$

also $I = neAv_d$, thus:

$\boxed{d\vec{F} = I\,d\vec{\ell} \times \vec{B}}$. Here $d\vec{\ell}$ has magnitude $d\ell$



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and direction same as that of the current density (or opposite to the drift of electrons).

FORCE ON ARBITRARY CURRENT CARRYING WIRE IN UNIFORM MAGNETIC FIELD:

Consider a current element $i\vec{dl}$ of an arbitrary shaped wire. Now magnetic force on the wire is given as:

$$d\vec{F} = i\vec{dl} \times \vec{B}, \text{ thus}$$

total force on the wire:

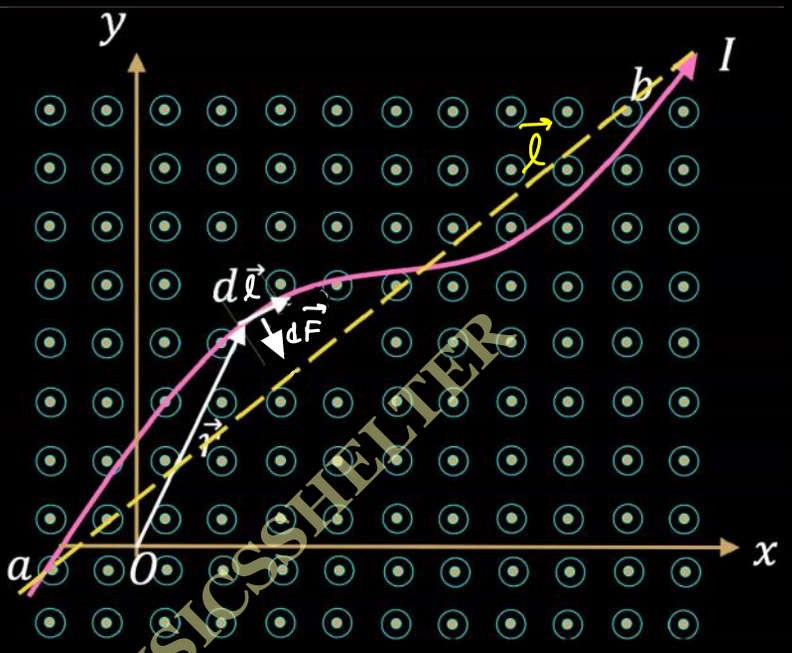
$$\vec{F} = \int i\vec{dl} \times \vec{B}$$

$$\Rightarrow \vec{F} = i\left[\int \vec{dl}\right] \times \vec{B}$$

($\because \vec{B}$ is uniform), Thus:

$$\boxed{\vec{F} = i\vec{l} \times \vec{B}}$$

Here $\vec{l}$ is the vector joining initial and final points of the current carrying wire.



→ For finding the magnitude of the force it is often convenient to use $F = i l_1 B$ or $F = i l B_1$ where l_1 represents component of $\vec{l}$ perpendicular to $\vec{B}$ and B_1 is component of $\vec{B}$ perpendicular to $\vec{l}$.

→ For a complete loop in uniform magnetic field the net force is zero because $\vec{l} = 0$ for complete loops.

→ Net force is independent of exact shape of the wire and only depends on initial and final points of the wire.

→ For straight thin wires in uniform magnetic fields, the effective point of application of magnetic force can be considered at the mid point of the wire because of uniform distribution of force.

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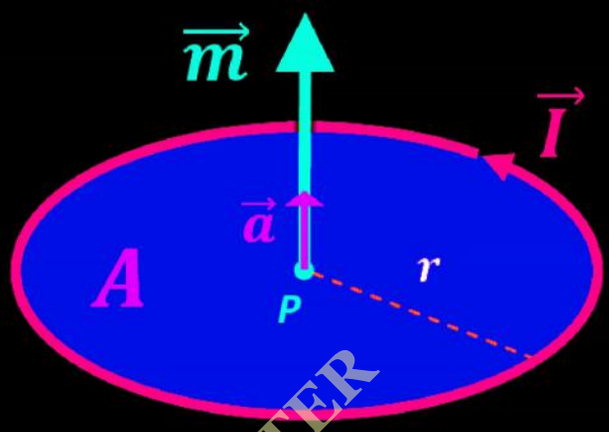
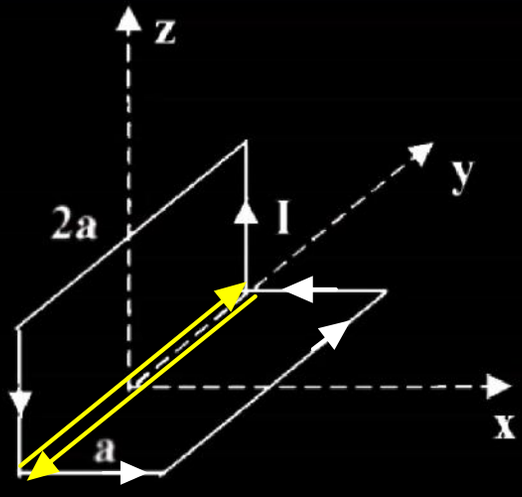
Magnetic Moment Of A Current Loop: In many expressions involving forces and torques on current carrying loops and magnetic fields generated by them, it is convenient to define the physical quantity "Magnetic Moment". If a current loop carries a current I, then magnetic moment of the loop is defined as



$$\boxed{\vec{m} = I\vec{A}}$$ where $\vec{A}$ is the area vector defined by right hand thumb rule, i.e curl the fingers of the right hand along the current, then the thumb gives the direction of $\vec{A}$, magnitude of $\vec{A}$ is equal to area enclosed by the loop.

→ For three dimensional loops, magnetic moment is found by breaking it in to several planar loops by imagining equal and opposite currents to form planar loops. For instance in the adjacent figure, we have imagined equal and opposite currents along yellow arrows to form two planar loops in the x-y and y-z planes. Thus net magnetic moment is $(2a^2\hat{i} + 2a^2\hat{k})\,I$



→ For multi turn coils, magnetic moment of each turn is added to get the net magnetic moment.

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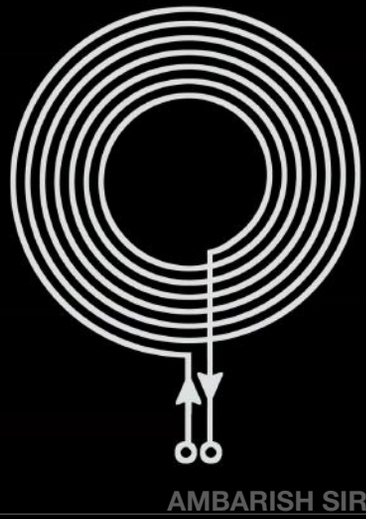
Example: Irodov 3.245:

A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8$ mA. The radii of inside and outside turns are equal to $a = 50$ mm and $b = 100$ mm. Find the magnetic moment of the spiral with a given current.



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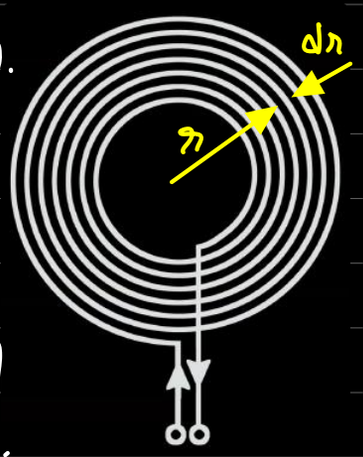
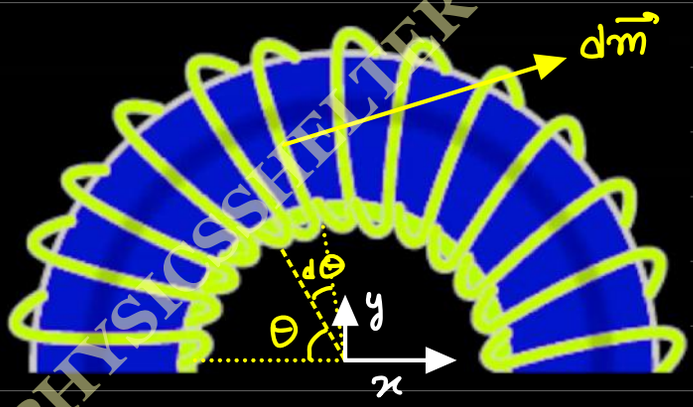
<u>Solution:</u> Consider a thickness dr of the spiral at a radius r. Number of turns per unit thickness is $N/(b-a)$. Thus number of turns in this thickness must be $[N/(b-a)]\,dr$. Thus elemental magnetic moment: $dm = \frac{N}{(b-a)} I\pi r^2 dr$, Now integrating this expression: $m = \frac{NI\pi}{(b-a)}\int_a^b r^2 dr = \frac{NI\pi}{3(b-a)}[b^3-a^3]$ Amp.



EXAMPLE: IRODOV 3.244:

Calculate the magnetic moment of a thin wire with a current I = 0.8 A, wound tightly on half a torus. The diameter of the cross-section of the tore is equal to d = 5.0 cm, the number of turns is N = 500.



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<u>Solution:</u> Consider an element of angular thickness $d\theta$ at angular position θ. For this element:

$$d\vec{m} = \left(\frac{N}{\pi}d\theta\right).\, I \times \frac{\pi d^2}{4}\left[\sin\theta\,\hat{i} + \cos\theta\,\hat{j}\right]$$

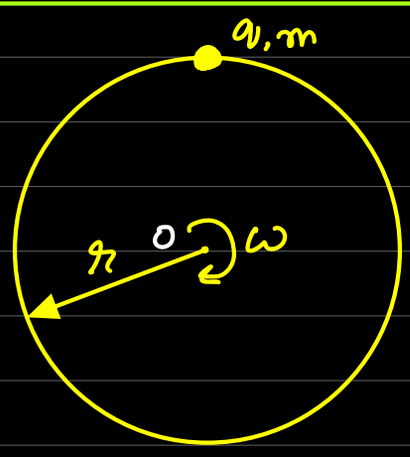
Thus total magnetic moment: $\vec{m} = \frac{NId^2}{4}\left[\hat{i}\int_0^{\pi}\sin\theta + \hat{j}\int_0^{\pi}\cos\theta\right]$

$$\Rightarrow \vec{m} = \frac{NId^2}{2}\hat{i} \text{ Amp.}$$

<u>ALITER:</u> Consider a thickness dl at angular location θ, for this: $d\vec{m} = \frac{N}{L}d\vec{l} \times \frac{\pi d^2}{4} I$ where L is total length of the tore (i.e Semicircle) and $d\vec{l}$ has the tangential direction. Thus: $I\frac{N\pi d^2}{4}\left(\frac{\vec{l}}{L}\right)$ where $\vec{l}$ is vector from initial to final point.

Now $\vec{l}/L = 2r/\pi r = 2/\pi$. Thus $\vec{m} = NId^2/2$ Amp.

Magnetic Moment To Angular Momentum Ratio Of A Spinning Charge:



Consider a charged particle of mass m and charge q spinning with an angular speed ω about an axis O (see figure). The equivalent current for such a spinning charge, $i = q\omega/2\pi$.

Thus:

$$\vec{m} = \frac{q\vec{\omega}}{2\pi} \times \pi r^2 = q\vec{\omega} r^2/2$$

Angular momentum of the particle about axis O: $\vec{L} = mr^2\vec{\omega}$

The ratio of magnetic moment to angular momentum

$$\boxed{\gamma = q/2m}$$

→ This factor γ is known as gyromagnetic ratio. The idea of gyromagnetic ratio can also be extended to uniformly charged spinning rigid bodies as will be seen in a following example.

Example: Irodov: 3.248:

A charge q is uniformly distributed over the volume of a uniform ball of mass m and radius R which rotates with an angular velocity ω about the axis passing through its centre. Find the respective magnetic moment and its ratio to the mechanical moment.

Solution: Let α be the charge of mass ratio of every particle of the sphere. Now magnetic moment of particle of charge q_i:

$$m_i = q_i(\omega/2\pi)\pi r_i^2 \quad \text{and its angular momentum:}$$

$$L_i = \frac{q_i}{\alpha} r_i^2 \omega = 2m_i/\alpha$$

Now applying summation on above equation:

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$$L = 2m/\alpha \;(1) \text{ or } m = \frac{QL}{2} = \frac{q}{2M}\cdot\left(\frac{2}{5}MR^2\omega\right)$$

$$\Rightarrow m = qR^2\omega/5 \text{ Am.}$$

→ Note that until equation (1), we nowhere used the fact that we are dealing with a solid sphere. The equation (1) can be conveniently use for finding magnetic moment of any arbitrary uniformly charged rigid body. $L = 2m/(Q/M) \Rightarrow \boxed{m = \frac{Q}{2M}L}$

→ As an example, suppose we had a spinning charged disc, $m = (q/2M)\cdot\frac{1}{2}MR^2\omega$ i.e. $qR^2\omega/4$ Am.

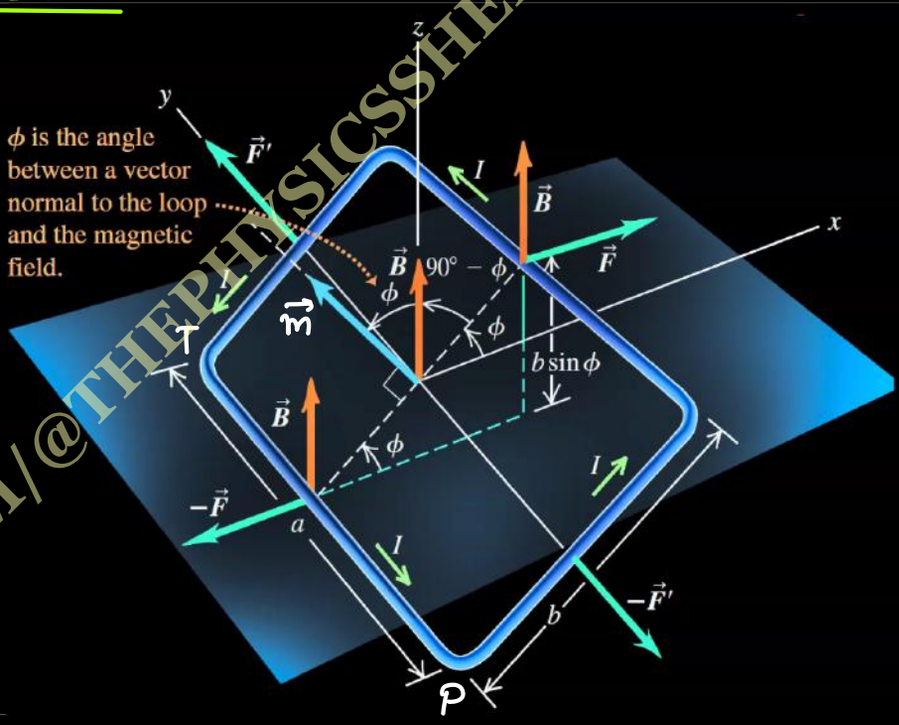
Magnetic Torque On A Current Loop:

Consider a current carrying loop of dimensions $a \times b$, such that $\vec{m}$ and $\vec{B}$ are inclined at an angle ϕ to each other as shown in the figure. It can be clearly seen that magnetic forces form a couple. The magnetic torque about any axis ∥ to PT is given as:

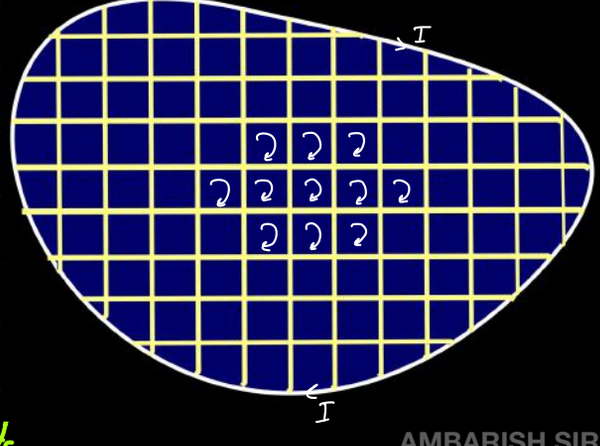
$$\tau = bF\sin\phi \times iaB$$

$\Rightarrow \tau = mB\sin(\phi)$, one can readily see that vectorially this torque is given as:

$$\boxed{\vec{\tau} = \vec{m} \times \vec{B}} \quad (1)$$



→ Even though we have proved this result for a rectangular loop, the same result also holds true for arbitrary loops. [This is because any arbitrary loop can be modeled as a collection of several rectangular loops. See figure, the elemental loop currents



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upon superposition effectively yield the current around the bigger loop boundary.

→ Recall that torque on an electric dipole in electric field is given as $\vec{\tau} = \vec{p} \times \vec{E}$, equation (1) has a very similar form and we can see that magnetic moment behaves just like electric dipole moment for purpose of finding torque in magnetic fields. This analogy holds not only for finding torque but also for finding several other physical quantities such as potential energy, and field due to point current loops (point magnetic dipole).

Potential Energy Of A Current Loop In Magnetic Field:

Just as we derived torque on an electric dipole by considering the work done by electric torque exactly in the same way we can obtain potential energy of a current carrying loop by considering the work done by magnetic torque. Thus magnetic potential energy (corresponding to "macroscopic" work done by magnetic field) of a current carrying loop is given as:

$$\boxed{U = -\vec{m} \cdot \vec{B}}$$

Magnetic Force On A Magnetic Dipole In Non Uniform Field:

Just as we derived electric force on an electric dipole, similar formulae hold true for forces on magnetic dipoles (current loops) in magnetic fields thus:

$$\boxed{\vec{F} = -\vec{\nabla} U = (\vec{m} \cdot \vec{\nabla})\vec{B}}$$

→ In case of electric dipole, $\vec{F} = (\vec{p} \cdot \vec{\nabla})\vec{E}$ was derived using the fact that $\vec{\nabla} \times \vec{E} = 0$, in case of magnetic fields, the equation is $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$ (Ampere's law), however in free space, $\vec{J} = 0$ and then $\vec{\nabla} \times \vec{B} = 0$ which is similar to $\vec{\nabla} \times \vec{E} = 0$.

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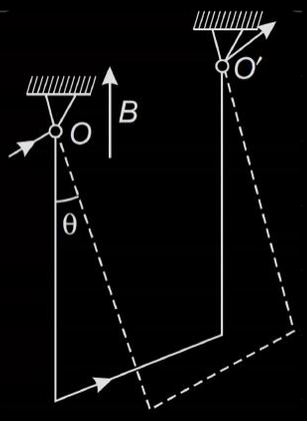
Example: Irodov: 3.253:

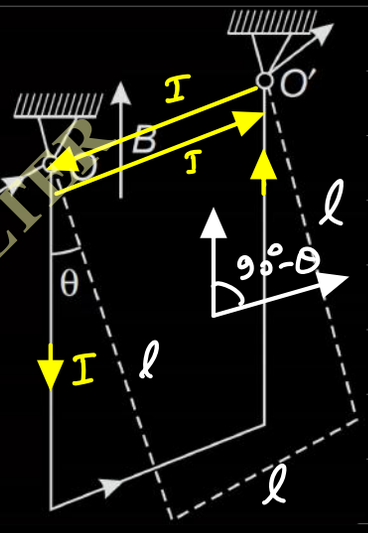
A copper wire with cross-sectional area $S = 2.5\ mm^2$ bent to make three sides a square can turn about a horizontal axis OO'. The wire is located in uniform vertical magnetic field. Find the magnetic induction if on passing a current $I = 16$ A through the wire the latter deflects by an angle $\theta = 20°$. (Density = ρ)



Solution: To conveniently find the torque we complete the loop by passing equal and opposite currents along OO'. Now torque due to complete loop about OO':

$\vec{\tau} = \vec{m} \times \vec{B} \Rightarrow |\vec{\tau}| = I\ell^2 B \cos\theta$, Torque due to additional line current is zero. Now magnetic torque must be balanced by gravitational torque. Gravitational torque:

$$\tau_g = \rho S \ell g (\ell \sin\theta) + \rho S \ell g \cdot \frac{\ell}{2} \sin\theta \times 2$$



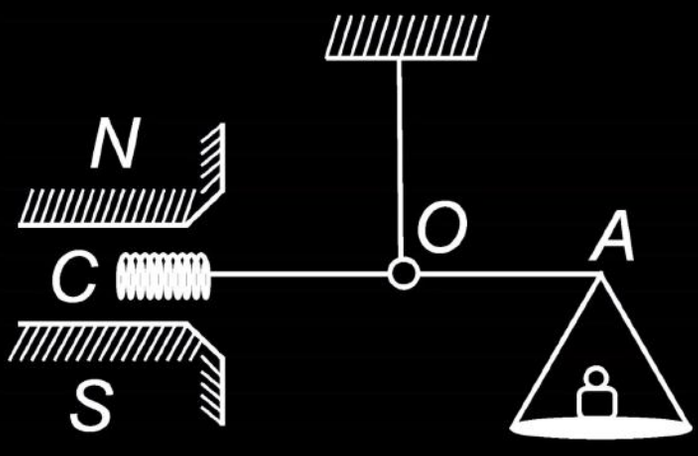
(Individual torques due to weight of each rod).

Thus $\tau_g = 2\rho S \ell^2 g \sin\theta = I\ell^2 B \cos\theta \Rightarrow B = 2\rho g S \tan\theta / I$ Ans.

Putting the values, $B = 10\ mT$.

Example: Irodov: 3.254:

A small coil C with $N = 200$ turns is mounted on one end of a balance beam and introduced between the poles of an electromagnet as shown in Fig. The cross-sectional area of the coil is $S = 1.0\ cm^2$, the length of the arm OA of the balance beam is $l = 30$ cm. When there is no current in the coil the balance is in equilibrium. On passing a current $I = 22$ mA through the coil the equilibrium is restored by putting the additional counterweight of mass $\Delta m = 60$ mg on the balance pan. Find the magnetic induction at the spot where the coil is located.



Solution: The key idea here is that torque due to a couple is independent of point. Thus magnetic torque about pivot O can be simply found using $\vec{\tau} = \vec{m} \times \vec{B}$ which is balanced due to gravitational torque. Thus:

$$NIS B \sin 90° = \Delta m g l \Rightarrow B = \frac{\Delta m g l}{NIS} = 0.4\ T \text{ Ans.}$$

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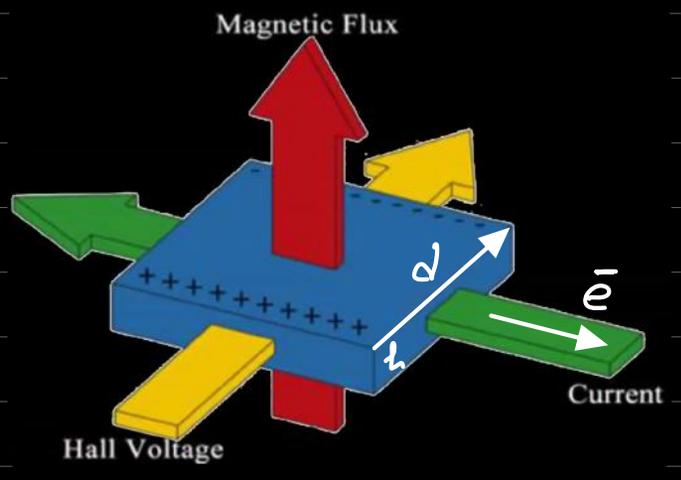
HALL EFFECT: At one point of time it was not well understood whether electric current in conductors is due to flow of positive charges or negative charges, Hall's experiment was a brilliant arrangement which demonstrated that the current was due to flow of electrons only. In this experiment, a cross magnetic field is apply to a current carrying cuboid, Now since the current is due to electron flow, due to magnetic forces, there is accumulation of negative charge on the rear side and a positive charge on the front side (see figure). Had the current been due to positive charge flows, rear face would have a build up of a positive charge instead of negative. Thus by putting a voltmeter across front and rear face, potential difference indicated that flow of current is due to movement of negative charges i.e. electrons. Quantitatively we can say that $eV_d B = eE$ where E is electric field from front face to the rear face. Thus $E = V_d B$, Now hall voltage:



$$V = Ed = V_d Bd, \quad \text{Now } I = nedh\,V_d$$

thus, $V = Bd \times \dfrac{I}{nedh} \Rightarrow \boxed{V = \dfrac{IB}{neh}}$

EXAMPLE: IRODOV: 3.267:

In Hall effect measurements in a sodium conductor the strength of a transverse field was found to be equal to $E = 5.0\ \mu V/cm$ with a current density $j = 200 A/cm^2$ and magnetic induction $B = 1.00$ T. Find the concentration of the conduction electrons and its ratio to the total number of atoms in the given conductor.

Solution: Here transverse field means field corresponding to Hall Voltage. Now in steady state, transverse magnetic force is balanced by transverse electric force. Thus $Ee = eV_d B \Rightarrow Ee = eB.\dfrac{I}{ne}$

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$\Rightarrow n = JB/Ee = 2.5 \times 10^{28}/m^3$, this is same as number of atoms per unit volume of a sodium crystals (can be derived using density data and mole concept). Thus required ratio: 1:1 Ans.

Example: Irodov: 3.268:

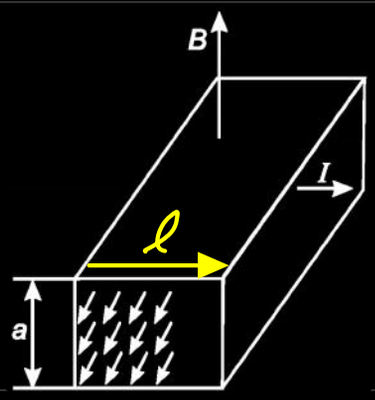
Find the mobility of the conduction electrons in a copper conductor if in Hall effect measurements performed in the magnetic field of induction $B = 100$ mT, the transverse electric field strength of the given conductor turned out to be $\eta = 3.1 \times 10^3$ times less than that of the longitudinal electric field. (Mobility is defined is drift velocity per unit longitudinal electric field).

Solution: Mobility $\mu = v_d/E_L$, Now $ev_dB = eE_t = e\frac{E_L}{\eta}$

$\Rightarrow \mu = v_d/E_L = 1/\eta B$ Ans ($\mu = 3 \times 10^{-2}\ m^2/Vs$ on putting values).

Example: Irodov 3.261:

In an electromagnetic pump designed for transferring molten metals a pipe section with metal is located in a uniform magnetic field of induction B. A current I is made to flow across this pipe section in the direction perpendicular both to the vector B and to the axis of the pipe. Find the gauge pressure produced by the pump if B = 0.10 T, I = 100 A, and a = 2.0 cm.



Solution: Here by gauge pressure we mean magnetic force per unit cross section area of the pipe that is responsible for pumping molten metal. Now $F = ilB$ and $A = al$ thus gauge pressure produced: $\Delta p = F/A = iB/a$ Ans.

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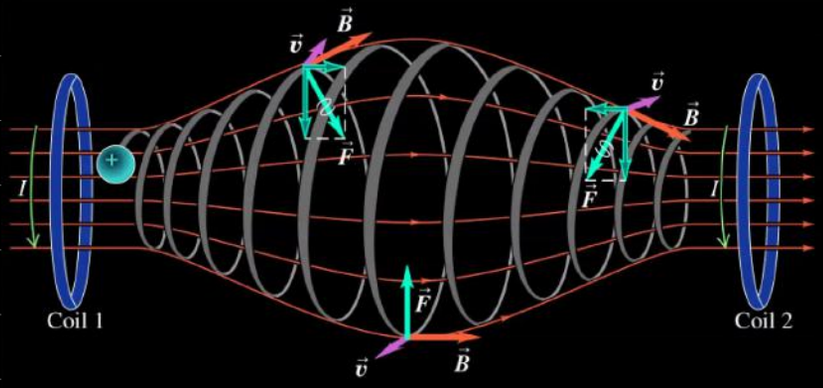
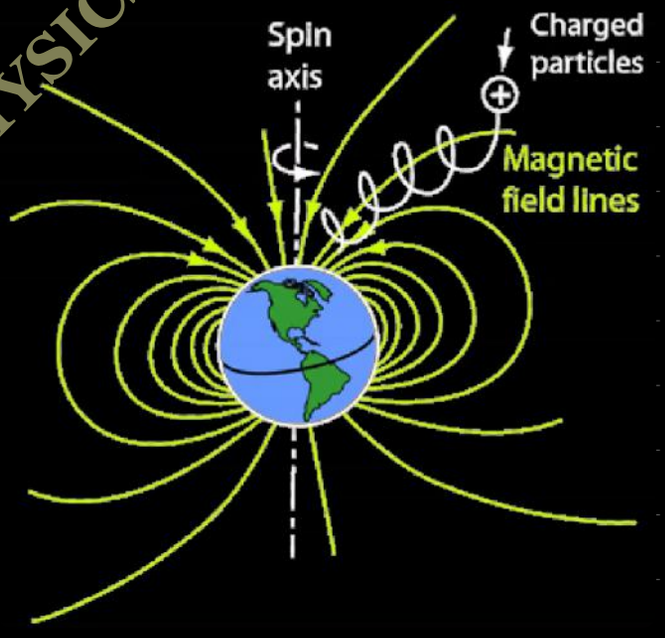
MAGNETIC BOTTLE / MIRROR:

For many scientific and engineering projects, it is important to confine high energy charged particles in a certain region of space (for example confining plasma for nuclear reactions). Magnetic fields can be effectively used for such confinement, as an example, a magnetic bottle / mirror arrangement is shown in the figure above where a charged particle keeps spiralling back and forth between two coils producing very strong magnetic fields.



AURORA BOREALIS (NORTHERN LIGHTS):



Numerous charged particles reach earth's atmosphere due to emissions from sun (Solar winds), these charges get trapped in earth's magnetic field and get concentrated near the poles due to structure of earth's field.

Near the poles, many of these particles have sufficient energy to ionize oxygen and nitrogen and thus putting them in excited states, during the de-excitation, these atoms emit light which create the beautiful phenomenon of Northern lights or Aurora Borealis.

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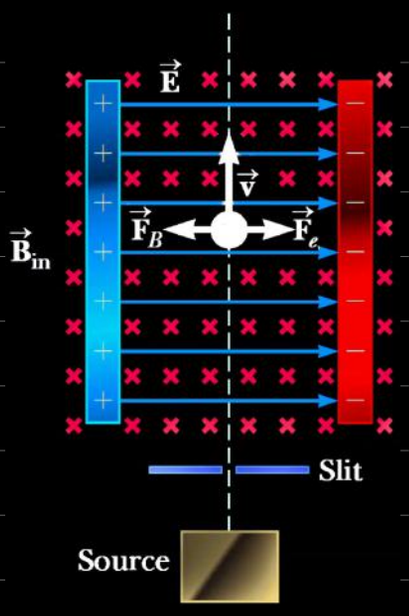
Velocity Selector: For certain scientific experiments, we need to have all charged particles to be moving with same velocity. This can be done using crossed electric and magnetic fields. Only particles for which $qE = qvB$ will be able to move straight and all other particles will be deflected side ways thus all particle escaping the velocity selector equipment will have velocity $v = E/B$. By suitably adjusting this ratio we can select the charged particles with certain desired velocities.

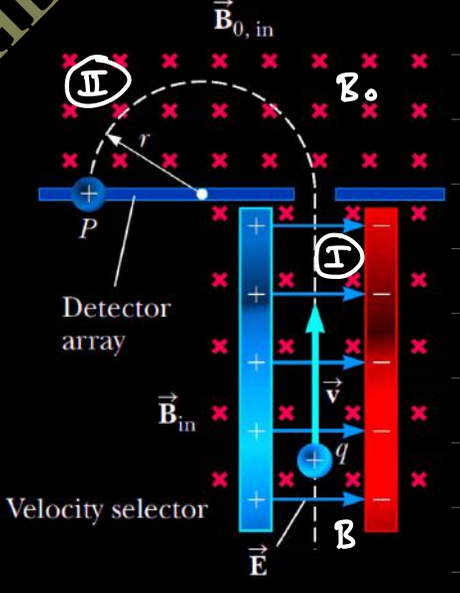


Mass Spectrometer: A mass spectrometer is a device which is used to separate ions according to their q/m ratio. It consists of two regions, in the first region, particles go along a straight line due to Lorentz force, i.e $\vec{E} + \vec{v} \times \vec{B} = 0$

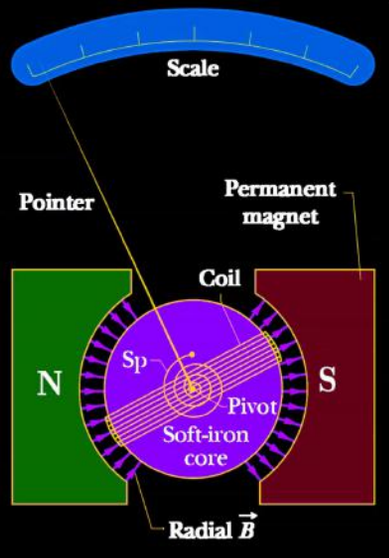
Now consider the adjacent figure, $v = E/B$, now in the second region (II), $r = mv/qB_0$

$$\Rightarrow r = \frac{m}{qB_0} \cdot \frac{E}{B} \Rightarrow r = \frac{Em}{qBB_0} \text{ or}$$

$$\boxed{q/m = E/rBB_0}$$ thus by knowing the location of point P (on the radius of path followed by ion), we can easily find the q/m ratio of any charged particle. This can be used to find composition of various isotope in an ion mixture.



Moving Coil Galvanometer: We introduced moving coil galvanometer in our discussions on electrical measuring instruments. Here we will try to understand its construction and working. A moving coil galvanometer consists of a rectangular coil of several turns wound on a soft iron cylindrical core. This coil is fixed on a pivot free to move in a radial magnetic field created by the



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poles of a magnet. A torsional spring is connected to the coil such that it provides a restoring torque when the coil is rotated from its equilibrium position. There is a pointer attached to the coil which can move on an etched dial to indicate the current passing through it. When a current passes through the coil, it experiences a magnetic torque:

$$\tau_M = NIAB\sin(90^\circ)$$

($\because m = NIA$, where N is number of turns, A is area of the coil and I is the current through the coil). Note that factor of $\sin 90^\circ$ appears in the cross product $\vec{m} \times \vec{B}$ because the field is radial (i.e always in plane of the coil and hence $\perp$ to its area vector).

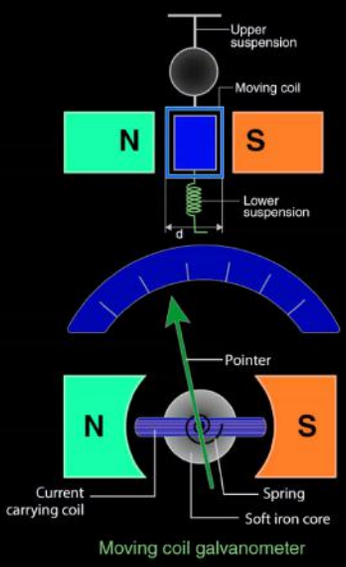
Now for equilibrium position of the coil, this magnetic torque is balanced by the restoring torque provided by the torsional spring (i.e $C\theta$).

Thus:

$$NIAB = C\theta \quad (C \text{ is torsional constant}, \theta \text{ is deflection angle})$$

Thus current is measured as:

$$\boxed{I = C\theta/(NAB)}$$



Moving coil galvanometer

CURRENT SENSITIVITY OF GALVANOMETER (S_i): The change in deflection angle of the galvanometer coil per unit change in current is called its current sensitivity. Thus:

$$S_i = d\theta/dI = NAB/C$$ Thus:

$$\boxed{S_i = NAB/C}$$

→ Current sensitivity can be improved either by increasing number of turns, increasing coil area, using a stronger magnetic field or using a less stiffer torsional spring (i.e reducing C).

→ Increasing the sensitivity will reduce the range of the galvanometer.

→ Sensitivity is typically specified in units of divisions/Amperes.

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Voltage Sensitivity Of A Moving Coil Galvanometer (S_V):

Just like current sensitivity, voltage sensitivity of a Galvanometer is specified is as change in deflection per unit change in the voltage applied against Galvanometer. From previous section,

$I = C\theta/NAB$, Let the resistance of Galvanometer coil per turn be λ. Then total resistance $R = \lambda N$. Now voltage drop

$V = IR \Rightarrow$

$$V = (C\theta/NAB) \times \lambda N = C\lambda\theta/AB.$$

Thus $S_V = d\theta/dV$

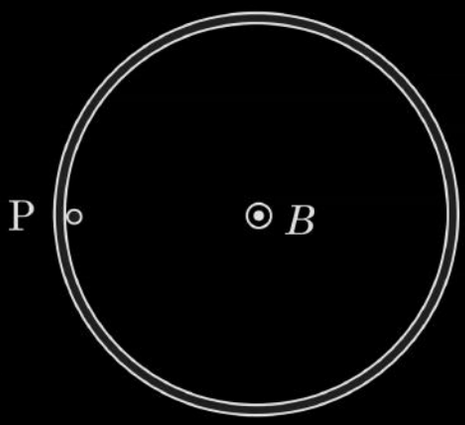
$$\Rightarrow \boxed{S_V = AB/C\lambda}$$

→ Thus voltage sensitivity can't be improved by increasing number turns (unlike current sensitivity) because voltage sensitivity is independent of N.

→ Voltage sensitivity can be improved by using coil of larger area, stronger magnet, less stiffen torsional spring and using a wire of smaller resistivity (or resistance per unit length).

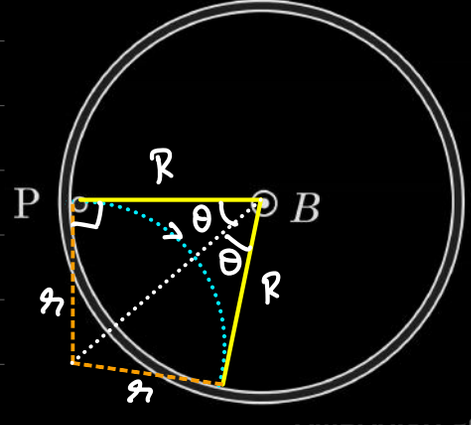
Example: Pathfinder: Build 8 (Altered):

In a hollow rigid cylinder of radius R gravity is absent and a uniform magnetic field of induction B is established everywhere parallel to the axis of the cylinder. The cylinder is made of an insulating material and is fixed by some external means. A very small ball of mass m and charge q is projected normally from a point P on the inner surface of the cylinder. If the ball returns to the point P after a certain number of elastic collisions with the cylinder, find possible velocities v of projection of the ball. Also find the possible velocities if the ball has to return to P after going only once around the center of the cylinder.



Solution: The initial path of the particle is shown in the adjacent figure and after every collision, same pattern repeats. We will first find the angle covered around the centre (i.e 2θ) between two successive collisions. From the figure:

$\tan\theta = r/R = (mv/qBR)$, Now for particle to return to initial point, some multiple of 2θ must be exactly an integer multiple of



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2π. Thus : $2\tan^{-1}(mv/qBR) \times n = 2k\pi$ ($k, n \in$ Natural No.).

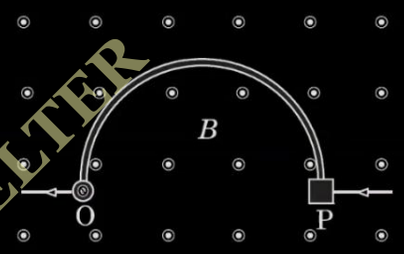
$v = \frac{qBR}{m}\tan(k\pi/n)$ Ans. Now for going only once around, $k=1$ Thus,

$v = \frac{qBR}{m}\tan(\pi/n)$, Note that maximum finite speed limit will occur for $n=3$

(∵ for $n=2$, $v=\infty$ and $\tan\pi/n$ monotonically decreases as n increases)

Example : Pathfinder: Build 24:

A schematic model of a current limiter consists of a conducting half ring of radius r and mass m, one end O of which is hinged while the other rests on a fixed metallic support P. The ring can rotate in a vertical plane without friction in presence of a uniform horizontal magnetic field of induction B pointing perpendicular to the plane of the ring. The ring is connected in series in a branch of a circuit through the hinge O and the metallic contact P. Find the minimum current, at which the limiter opens the circuit. The acceleration of free fall is g.



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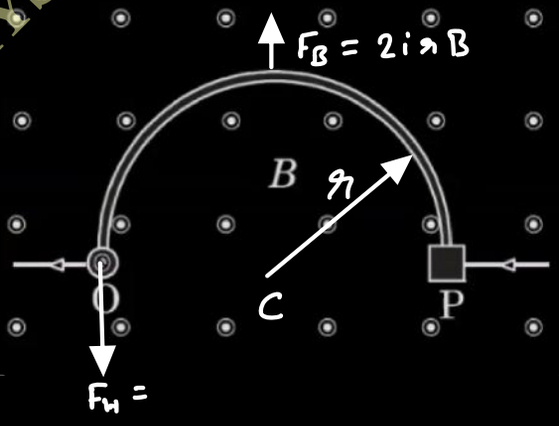
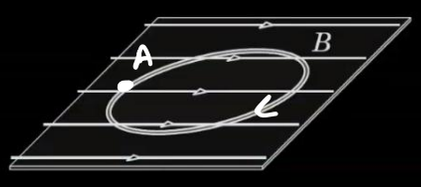
Solution: In the limiting case, normal reaction at P will go to zero and at that instant, Reaction at O must balance the resultant of mg and magnetic force. Thus: $F_H = 2inB - mg$. Now recognise that magnetic force at each element passes through the centre C. Now balancing torque about C: $F_H \times R = 0$ (∵ torque of mg about C is also zero)

Thus $F_H = 0 \Rightarrow 2inB = mg \Rightarrow i = mg/2nB$ Ans.



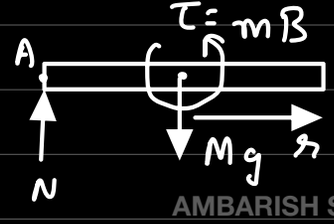
Example: Pathfinder: Build-27:

A uniform rigid ring of radius r and unknown mass is placed on a nonconducting frictionless horizontal tabletop, where exists a uniform horizontal magnetic field of induction B as shown in the figure. A current I that is double of the minimum current required to flip over the ring is made to circulate in the coil. Find force of normal reaction from the tabletop on the ring immediately after the current is switched on.



Solution: $I/2$ is the minimum current for flipping the ring. Now balancing the torques in limiting case, $Mg \times r = mB = \frac{I}{2} \times \pi r^2 B$ (1)

$\Rightarrow Mg = \frac{\pi I}{2} Br$ (2)



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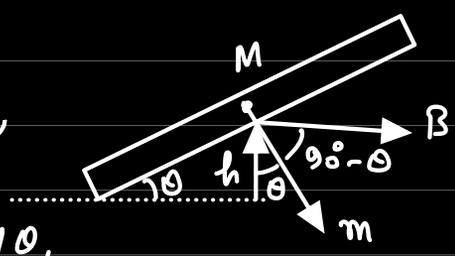
Note that if torques are just balanced, the ring will just flip over as can be inferred from energy consideration. When the loop has rotated through an angle θ,



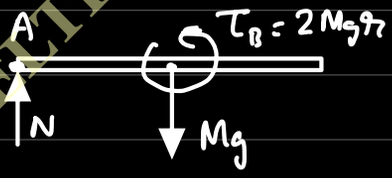
Gain in gravitational PE: $\Delta U_G = Mgr\sin\theta$,

Loss in Magnetic PE: $-\Delta U_B = mB\sin\theta$

But from (1), $mB = Mgr$, thus at a general θ,

Gain in Gravitational PE is exactly equal to loss in magnetic PE, thus, the loop can rise with negligible KE is current is very slightly greater than $I/2$.

Now when a current I flows, The magnetic torque about point A will become double the torque due to gravity.



Now using the torque equation about axis through A:

$$MgR = \frac{3}{2}MR^2\alpha \quad (3)$$

(Note that since A slides horizontally, torque of pseudo force is zero).

Now for vertical direction:

$$N - Mg = MR\alpha \quad (4)$$

From (2), $Mg = \pi IBr/2$ and from (3), $MR\alpha = \frac{2}{3}Mg = \frac{2}{3}\times\frac{\pi IBr}{2}$, thus

$$N - \pi IBr/2 = \pi IBr/3 \Rightarrow N = 5\pi IBr/6 \text{ ans.}$$

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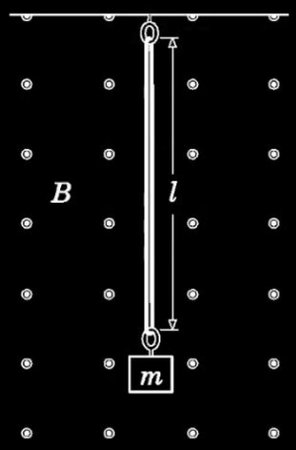
EXAMPLE: PATHFINDER: CHECK: 8:

A loop is made by joining the ends of a flexible, inextensible and light conducting wire of length $2l$ with a thin insulating coating. The loop passes through two frictionless hooks, one of the hooks is attached to the ceiling and from the other hook a small rubber block of mass is suspended as shown in the figure. Below the ceiling, a horizontal magnetic field of induction B is established. Now a gradually increasing current is made to circulate in the loop. For all values of current in the loop, forces of magnetic interaction between different parts of the wire are negligible as compared to other forces.



(a) What shape will the wire loop assume for a particular value of current in the loop?

(b) Find the maximum height up to which the block can be lifted.

(c) Find the current in the loop to lift the block to a height $h = l\left(1 - \frac{3}{\pi}\right)$.

Solution: Since it is a massless wire and magnetic force is everywhere perpendicular to wire, for tangential force balance, tension must

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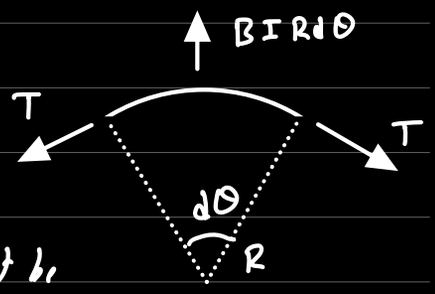
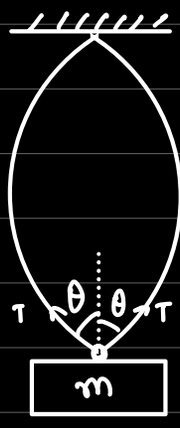
be same throughout the wire. We will show that under such circumstances, the wire must assume the shape of arc of a circle.



Now consider FBD of small portion of the wire,

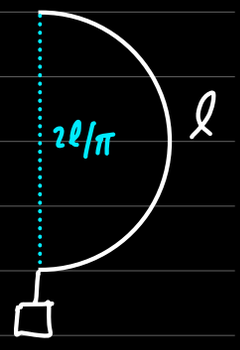
by force balance, $2T\sin(d\theta/2) = BIRd\theta$

$\Rightarrow R = T/BI$ (1), Now Since T, B and I are constant over each element of the wire $\therefore$ R must be constant which means that wire must assume the shape of arc of a circle. Now Considering the FBD of the block, $2T\cos\theta = mg$ (2), Now as the block rises, θ increases thus $\cos\theta$ decreases and Tension accordingly must increase ($\because$ (2)),



The least radius of curvature occurs when the arcs bulge like semi-circles (Reflex arc is not possible because it will make the tension downward which can't balance mg). Thus maximum height :

$h_{max} = l - 2l/\pi = l(1 - 2/\pi)$ Ans.

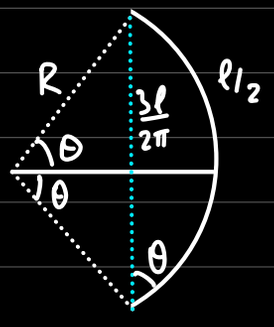


(c) The lifted height $h = l(1 - 3/\pi) = l$ - chord length

$\Rightarrow$ chord length $= 3l/\pi$. Now:

$R\sin\theta = 3l/2\pi$ and $R\theta = l/2 \Rightarrow \sin\theta = 3\theta/\pi$

This is a transcendental equation, though its solution can be guessed. By a guess, $\theta = \pi/6$, thus $R = l/2\theta = 3l/\pi$.



Thus from (1) $\frac{3l}{\pi} = \frac{T}{BI}$ (4) and from (2), $T\sqrt{3} = mg$ (5)

From (4) & (5), Now, $I = \pi mg/3\sqrt{3}Bl$ Ans.

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